



# TRIAL EXAMINATION 2025

## FORM VI CHEMISTRY

### General Instructions

- Reading Time: 5 minutes
- Working Time: 3 hours
- Write using black pen

### Structure of Paper & Instructions

- Section 1: Multiple-choice (20 marks).  
Spend 35 minutes on this section.
- Section 2: Written answer (80 marks).  
Spend 2 hours 25 minutes on this section.

### Exam Instructions

- Remove the centre staple and hand in all parts of the paper in a neat bundle.
- Show all relevant working.
- **Write your candidate number in the space provided on pages 1, 11, 17, 23 and 27.**

**Total marks: 100**

**Friday 8 August**  
**12:00 pm**

### Checklist

Each boy should have the following:

- ☐ 1 question paper
- ☐ 1 multiple-choice answer sheet
- ☐ 1 data sheet

JL Senczyszyn

Examiners: EJS, JAC, JLS, TW

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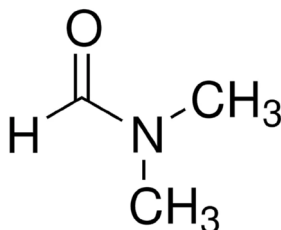
## SECTION I: MULTIPLE CHOICE QUESTIONS

### 20 marks

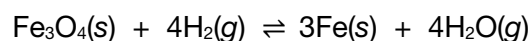
Attempt ALL Questions  
Use the Multiple-Choice Answer Sheet.

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- 1 What is the IUPAC name for the structure shown?



- (A) *N*-dimethylmethanal  
(B) *N*-ethylmethanamide  
(C) *N,N*-dimethylmethanamine  
(D) *N,N*-dimethylmethanamide
- 2 Which of the following compounds is produced in the reaction between hydrochloric acid and calcium carbonate but **not** in the reaction between hydrochloric acid and calcium hydroxide?
- (A) calcium chloride  
(B) carbon dioxide  
(C) hydrogen  
(D) water
- 3 What is the equilibrium expression for the equilibrium system shown?



- (A)  $K_{eq} = \frac{1}{[\text{H}_2]^4}$   
(B)  $K_{eq} = \frac{[\text{H}_2\text{O}]}{[\text{H}_2]}$   
(C)  $K_{eq} = \frac{[\text{H}_2\text{O}]^4}{[\text{H}_2]^4}$   
(D)  $K_{eq} = \frac{[\text{Fe}]^3[\text{H}_2\text{O}]^4}{[\text{Fe}_3\text{O}_4][\text{H}_2]^4}$

4 Which of the following substances is an Arrhenius base?

- (A)  $\text{Li}_2\text{CO}_3$
- (B)  $\text{NH}_3$
- (C)  $\text{NaHCO}_3$
- (D)  $\text{Sr}(\text{OH})_2$

5 Which compound listed below will react in the presence of concentrated sulfuric acid with the product of its own oxidation to produce an ester?

- (A) propan-1-ol
- (B) propan-2-ol
- (C) propanal
- (D) propanoic acid

6 Which compound will produce a peak at 210 ppm in  $^{13}\text{C}$  NMR spectroscopy?

- (A) pentan-1-ol
- (B) pentan-1-amine
- (C) pentanoic acid
- (D) pentanal

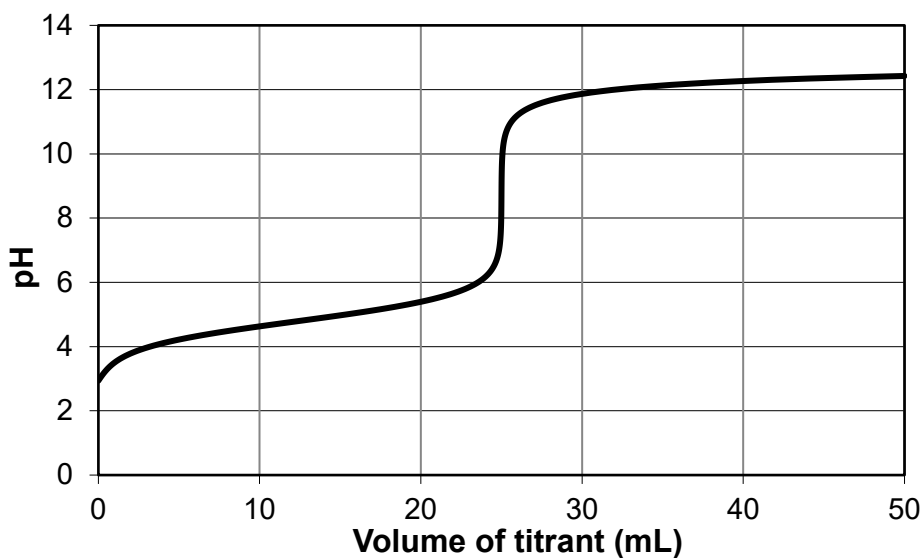
7 Consider the equilibrium system shown for the production of nitrosyl chloride (NOCl).



Which of the following conditions will maximise the equilibrium yield of nitrosyl chloride?

|     | <i>Pressure</i> | <i>Temperature</i> |
|-----|-----------------|--------------------|
| (A) | high            | high               |
| (B) | high            | low                |
| (C) | low             | high               |
| (D) | low             | low                |

Use the information below to answer questions 8 and 9.



8 Which of the following titrations would result in this titration curve?

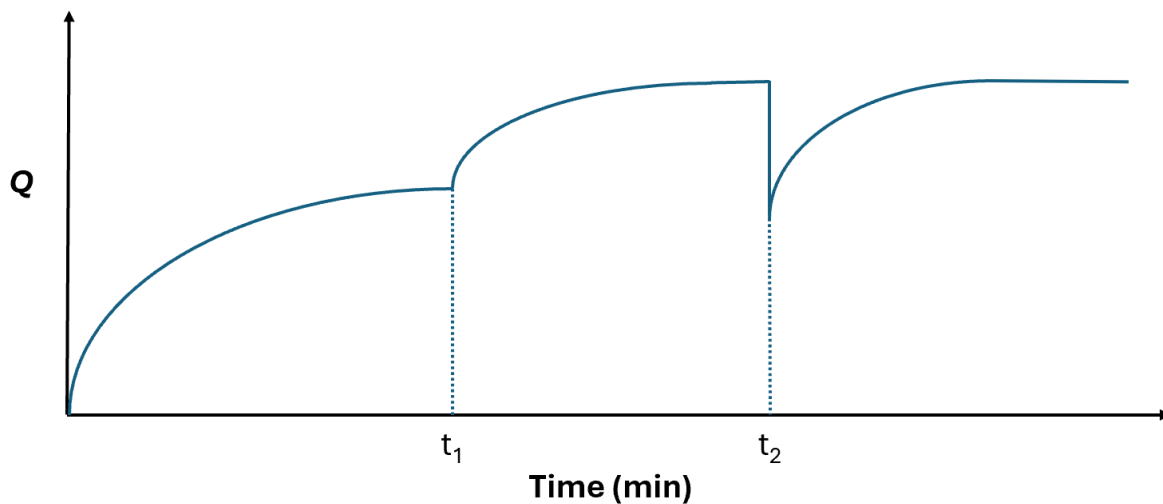
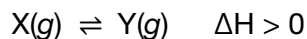
- (A)  $0.1 \text{ mol L}^{-1} \text{ C}_2\text{H}_5\text{COOH}$  with  $0.1 \text{ mol L}^{-1} \text{ NH}_3$
- (B)  $0.1 \text{ mol L}^{-1} \text{ C}_2\text{H}_5\text{COOH}$  with  $0.1 \text{ mol L}^{-1} \text{ NaOH}$
- (C)  $0.1 \text{ mol L}^{-1} \text{ HCl}$  with  $0.1 \text{ mol L}^{-1} \text{ NaOH}$
- (D)  $0.1 \text{ mol L}^{-1} \text{ HCl}$  with  $0.1 \text{ mol L}^{-1} \text{ NH}_3$

9 Which of the following indicators is most suitable for determining the endpoint of the titration above?

|     | <i>Indicator</i>  | <i>Colour change pH range</i> |
|-----|-------------------|-------------------------------|
| (A) | Methyl orange     | 3.1 – 4.4                     |
| (B) | Bromothymol blue  | 6.0 – 7.6                     |
| (C) | Phenolphthalein   | 8.2 – 10.0                    |
| (D) | Alizarin yellow R | 10.1 – 12.0                   |

- 10** An aqueous solution containing which substance is most likely to have a pH of 9?
- (A) methanol
  - (B) propanoic acid
  - (C) ethanamine
  - (D) butanal
- 11** In proton NMR, what do the relative peak areas indicate about a pure substance?
- (A) The number of protons in the molecule.
  - (B) The ratio of protons in different environments.
  - (C) The proximity of the proton to an electronegative element.
  - (D) The molecular weight of the molecule.
- 12** Infrared spectroscopy of a pure compound produces a broad peak centered at  $3415\text{ cm}^{-1}$ . Which series of organic compounds is the unknown compound likely to belong to?
- (A) alcohol
  - (B) carboxylic acid
  - (C) aldehyde
  - (D) alkene
- 13** Which of the following mixtures of aqueous species will form a buffer solution when both species have a concentration of approximately  $0.1\text{ mol L}^{-1}$ .
- (A)  $\text{HCl} / \text{Cl}^-$
  - (B)  $\text{HCOOH} / \text{HCOO}^-$
  - (C)  $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$
  - (D)  $\text{H}_2\text{O} / \text{OH}^-$

14 The graph shows changes in  $Q$  over time for the equilibrium system shown.



Which of the following could be the changes made at  $t_1$  and  $t_2$ ?

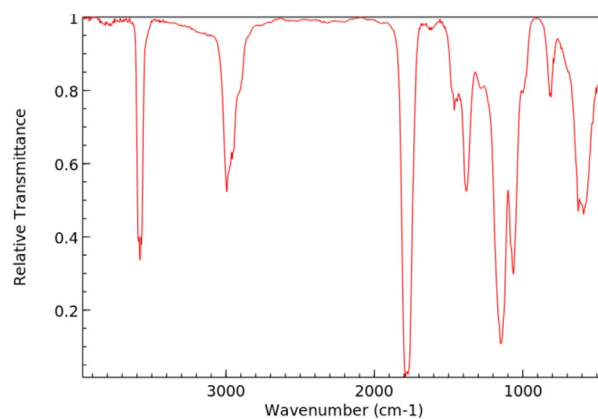
|     | <i>Change made at <math>t_1</math></i> | <i>Change made at <math>t_2</math></i> |
|-----|----------------------------------------|----------------------------------------|
| (A) | Decrease in temperature                | Addition of X                          |
| (B) | Decrease in temperature                | Removal of X                           |
| (C) | Increase in temperature                | Addition of Y                          |
| (D) | Increase in temperature                | Removal of Y                           |

15 Which set of peaks would be likely found in the mass spectrum of 2,2-dimethylpropanal?

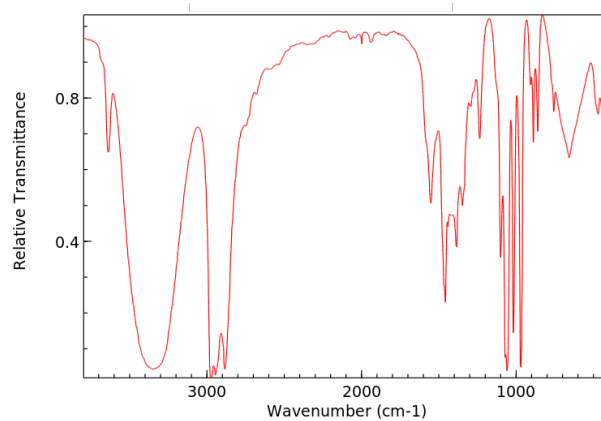
- (A)  $m/z = 14$ ,  $m/z = 28$ ,  $m/z = 57$ ,  $m/z = 71$  and  $m/z = 85$
- (B)  $m/z = 14$ ,  $m/z = 29$ ,  $m/z = 58$ ,  $m/z = 73$  and  $m/z = 87$
- (C)  $m/z = 15$ ,  $m/z = 29$ ,  $m/z = 57$ ,  $m/z = 71$  and  $m/z = 86$
- (D)  $m/z = 15$ ,  $m/z = 28$ ,  $m/z = 58$ ,  $m/z = 73$  and  $m/z = 86$

16 Which infrared spectrum was most likely produced from a pure sample of propanone?

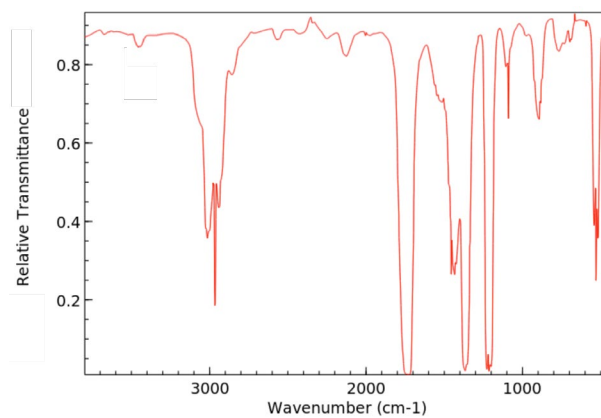
(A)



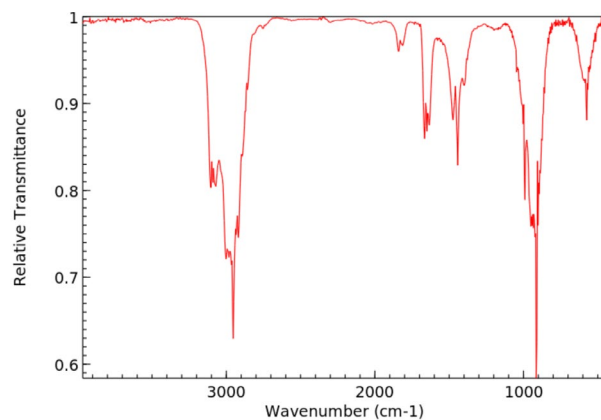
(B)



(C)



(D)



- 17** In an experiment to determine the  $K_{sp}$  value of barium fluoride, five 1.0 mL samples containing barium ions were prepared and each was added to a 1.0 mL solution containing fluoride ions. The results are given below.

| Sample | Barium ion solution                          |             | Fluoride ion solution                        |             | Observation             |
|--------|----------------------------------------------|-------------|----------------------------------------------|-------------|-------------------------|
|        | Initial concentration (mol L <sup>-1</sup> ) | Volume (mL) | Initial concentration (mol L <sup>-1</sup> ) | Volume (mL) |                         |
| 1      | 0.010                                        | 1.0         | 0.026                                        | 1.0         | No precipitate          |
| 2      | 0.010                                        | 1.0         | 0.028                                        | 1.0         | No precipitate          |
| 3      | 0.010                                        | 1.0         | 0.030                                        | 1.0         | Faint white precipitate |
| 4      | 0.010                                        | 1.0         | 0.032                                        | 1.0         | White precipitate       |

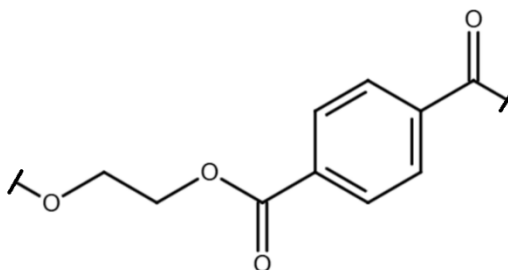
Which of the following could be the  $K_{sp}$  value of barium fluoride?

- (A)  $1.0 \times 10^{-6}$
  - (B)  $1.2 \times 10^{-6}$
  - (C)  $8.8 \times 10^{-6}$
  - (D)  $7.2 \times 10^{-5}$
- 18** When ethene is reacted completely with hydrogen bromide, 7.27 L of the product is obtained at 25.0°C and 120 kPa. What is the mass of ethene that was reacted?
- (A) 7.27 g
  - (B) 9.87 g
  - (C) 79.7 g
  - (D) 118 g

**19** A salt has a  $K_{sp}$  value of  $4.3 \times 10^{-10}$  and a molar solubility of  $2.0 \times 10^{-3} \text{ mol L}^{-1}$ . Which of the following represents the formula of this salt?

- (A) XY
- (B)  $\text{XY}_2$
- (C)  $\text{XY}_3$
- (D)  $\text{X}_2\text{Y}_3$

**20** 1.242 kg of a single polymer with the repeating unit shown below was obtained after a polymerisation reaction.



What mass of water is released during this polymerisation reaction?

- (A) 116 g
- (B) 233 g
- (C) 656 g
- (D) 803 g

## SECTION II: WRITTEN ANSWER QUESTIONS

**80 marks**

Attempt ALL Questions

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### Question 21 (2 marks)

**Marks**

Using balanced chemical equations, show that the  $\text{H}_2\text{PO}_4^-$  ion can react with both strong acids and strong bases.

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### Question 22 (3 marks)

Calculate the pH of  $0.20 \text{ mol L}^{-1}$  hydrazoic acid ( $\text{HN}_3$ ,  $K_a 1.9 \times 10^{-5}$ ).

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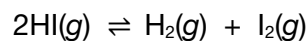
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**Question 23** (6 marks)**Marks**

Hydrogen iodide decomposes when heated to hydrogen and iodine as shown below.



- (a) Give the equilibrium expression for this reaction.

**1**

- (b) In an experiment, 1.75 mol of hydrogen iodide was placed in a 1.00 L vessel at a constant temperature. At equilibrium, the concentration of iodine gas was  $0.168 \text{ mol L}^{-1}$ .

What is the  $K_{eq}$  value at this temperature?

**3**

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- (c) Explain what would happen to the value of  $K_{eq}$  if a catalyst was added at constant temperature.

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**Question 24** (4 marks)

## Marks

100.0 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide is mixed with 100.0 mL of 1.00 mol L<sup>-1</sup> nitric acid in a cardboard cup. The initial temperature of each solution is 24.30 °C and the resulting solution has a temperature of 31.15 °C.

You can assume that there is no volume change upon reaction, and that the resulting solution has a density of  $1.04 \text{ g mL}^{-1}$  and a specific heat capacity of  $4.03 \text{ J K}^{-1} \text{ g}^{-1}$ .

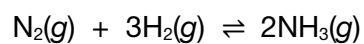
Calculate the enthalpy of neutralisation for this reaction.

4

[illegible]

**Question 25** (10 marks)**Marks**

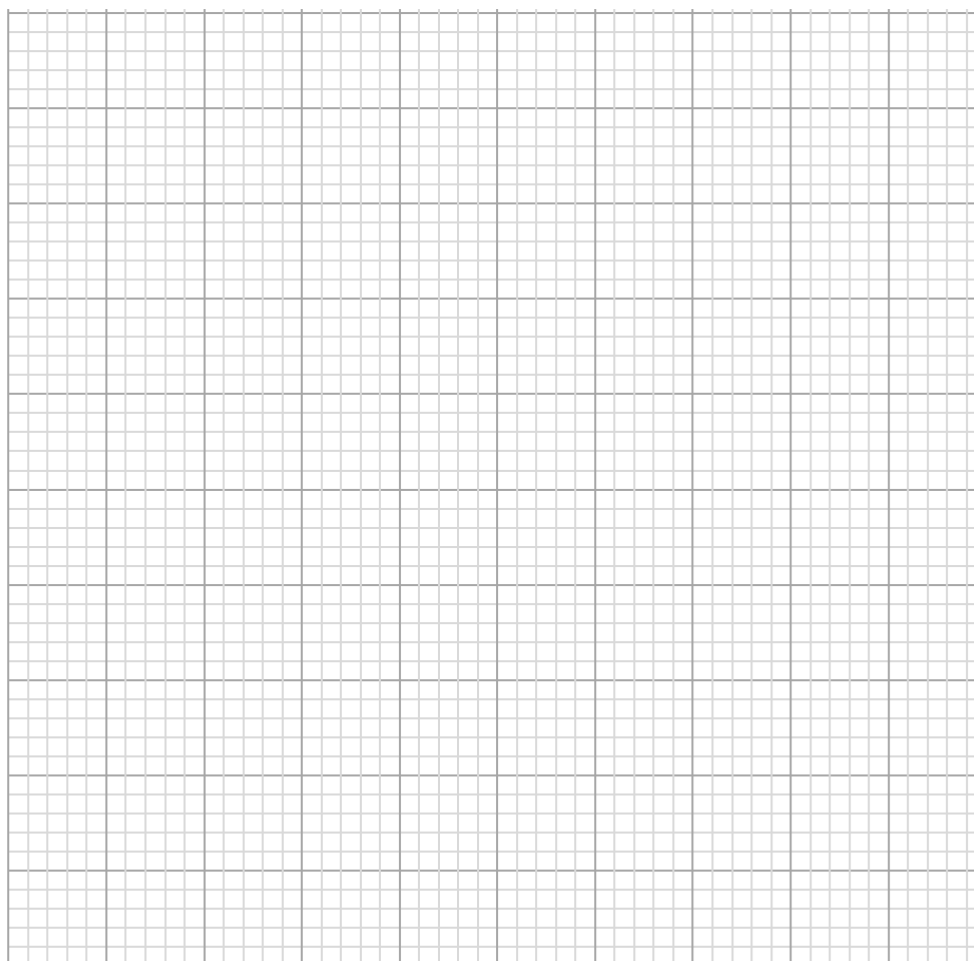
The synthesis of ammonia from its elements is an equilibrium process.



The table shows the value of  $K_{\text{eq}}$  for this reaction at a range of different temperatures.

| Temperature (K) | $K_{\text{eq}}$ |
|-----------------|-----------------|
| 390             | 90              |
| 400             | 44              |
| 420             | 12              |
| 430             | 6.4             |
| 440             | 3.6             |

- (a) Plot this data, including a curved line of best fit.

**3** $K_{\text{eq}}$ **Temperature (K)**

**Question 25 continued.****Marks**

- (b) Using the graph, estimate  $K_{eq}$  at 410 K.

**1**

- (c) A system at 410 K is found to have  $[N_2] = 0.24 \text{ mol L}^{-1}$ ,  $[H_2] = 1.1 \text{ mol L}^{-1}$  and  $[NH_3] = 1.6 \text{ mol L}^{-1}$ .

Show that this system is not at equilibrium and use  $Q$  and  $K_{eq}$  to explain how the system will shift to reach equilibrium.

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- (d) Is the production of ammonia exothermic or endothermic? Justify your answer using Le Châtelier's Principle and the data provided.

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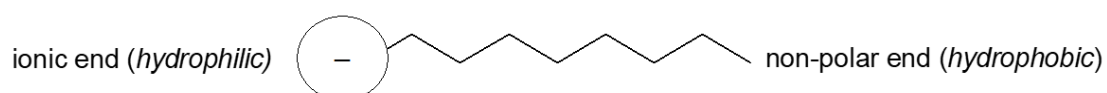
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**Question 26** (5 marks)**Marks**

Soap can be used as a cleaning agent due to its ability to stabilise emulsions.

(a) Draw a labelled diagram to show the structure of a soap molecule.

**2**

(b) Explain how soap can act as a cleaning agent.

**3**

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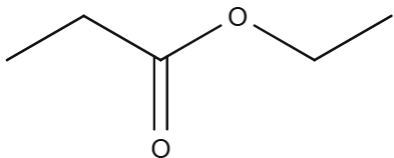
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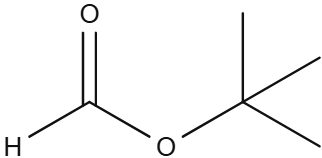
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**Question 27** (4 marks)

Two esters, A and B, have molecular formula  $C_5H_{10}O_2$ .

Proton NMR spectroscopy was performed on both compounds, and the following data was obtained.

| <b>Ester A</b>                                                                    |                             |                           |                               |
|-----------------------------------------------------------------------------------|-----------------------------|---------------------------|-------------------------------|
|  | <i>Chemical shift (ppm)</i> | <i>Relative peak area</i> | <i>Peak splitting pattern</i> |
|                                                                                   | 1.05                        | 3                         | triplet                       |
|                                                                                   | 1.16                        | 3                         | triplet                       |
|                                                                                   | 2.26                        | 2                         | quartet                       |
|                                                                                   | 4.11                        | 2                         | quartet                       |

| <b>Ester B</b>                                                                      |                             |                           |                               |
|-------------------------------------------------------------------------------------|-----------------------------|---------------------------|-------------------------------|
|  | <i>Chemical shift (ppm)</i> | <i>Relative peak area</i> | <i>Peak splitting pattern</i> |
|                                                                                     | 1.42                        | 9                         | singlet                       |
|                                                                                     | 9.60                        | 1                         | singlet                       |

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**Question 27 continued.**

### Marks

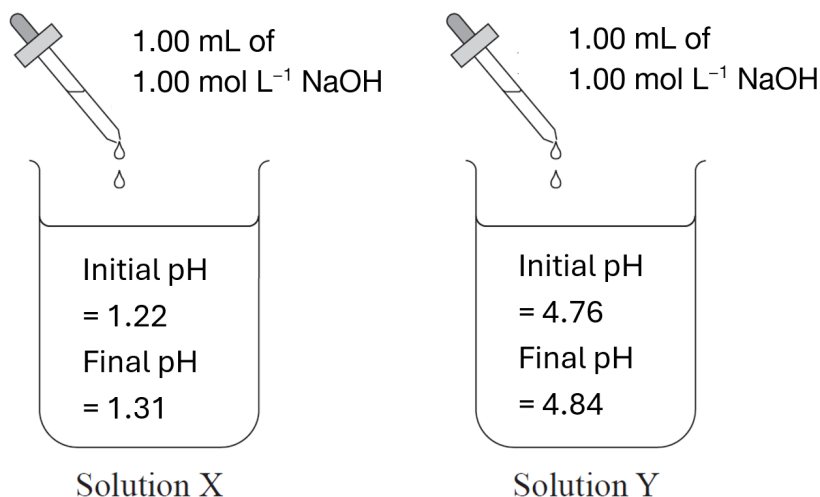
State **two** differences between the proton NMR data and relate these to the structure of the two esters.

4

[illegible]

**Question 28** (7 marks)**Marks**

The pH of two 100 mL solutions, X and Y, were measured before and after 1.00 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide solution was added to each.



One of these solutions contains a solution of a weak acid and its conjugate base, both at a concentration of 0.10 mol L<sup>-1</sup>; the other contains a 0.060 mol L<sup>-1</sup> solution of hydrochloric acid.

- (a) By performing a relevant calculation, show that Solution X is the 0.060 mol L<sup>-1</sup> solution of hydrochloric acid.

**2**

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- (b) Calculate the  $K_a$  of the weak acid.

**1**

**Question continues on next page.**

**Question 28 continued.**

### Marks

When 1.00 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide solution is added to 100 mL of water, the pH increases from 7.00 to 12.00.

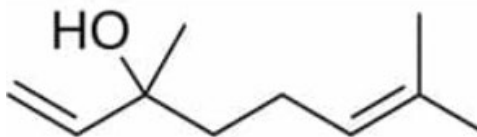
- (c) Explain why adding 1.00 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide solution to both solution X and solution Y does not change pH of either appreciably.

4

[illegible]

**Question 29** (5 marks)**Marks**

Linalool is a compound found in many flowers and spices. The structural formula is shown below.



- (a) State the names of the **two** functional groups in linalool.

**2**

- (b) Linalool undergoes many chemical reactions. Complete the table by identifying the product when the following reagents are added to a linalool solution under the stated conditions.

**3**

| <i>Reagent and conditions</i>                                                      | <i>Structural formula of organic product</i> |
|------------------------------------------------------------------------------------|----------------------------------------------|
| $\text{H}_2(\text{g})$<br>Pd catalyst                                              |                                              |
| $\text{Br}_2(\text{aq})$                                                           |                                              |
| $\text{HCOOH}(\text{aq})$<br>$\text{H}_2\text{SO}_4$ catalyst<br>Heat under reflux |                                              |

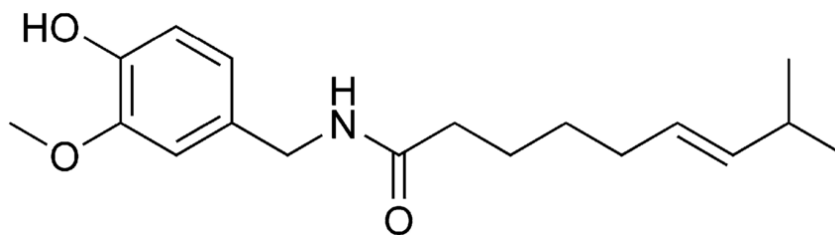
**Question 30** (3 marks)

(3 marks)

## Marks

The compound responsible for the hot taste of chillies, capsaicin, is shown. Drinking full cream milk, which contains fat, is recommended after eating meals containing lots of chilli as it reduces the hot taste on the tongue.

Explain why milk is better than water in removing capsaicin from the tongue.



3

[illegible]

**Question 31** (9 marks)**Marks**

Three bottles, known to contain pentan-1-ol, pentan-3-ol and 2-methylbutan-2-ol, are missing labels.

- (a) Suggest how a student could distinguish between the three alcohols using one or more chemical tests.

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**Question continues on next page.**

**Question 31 continued.**

## Marks

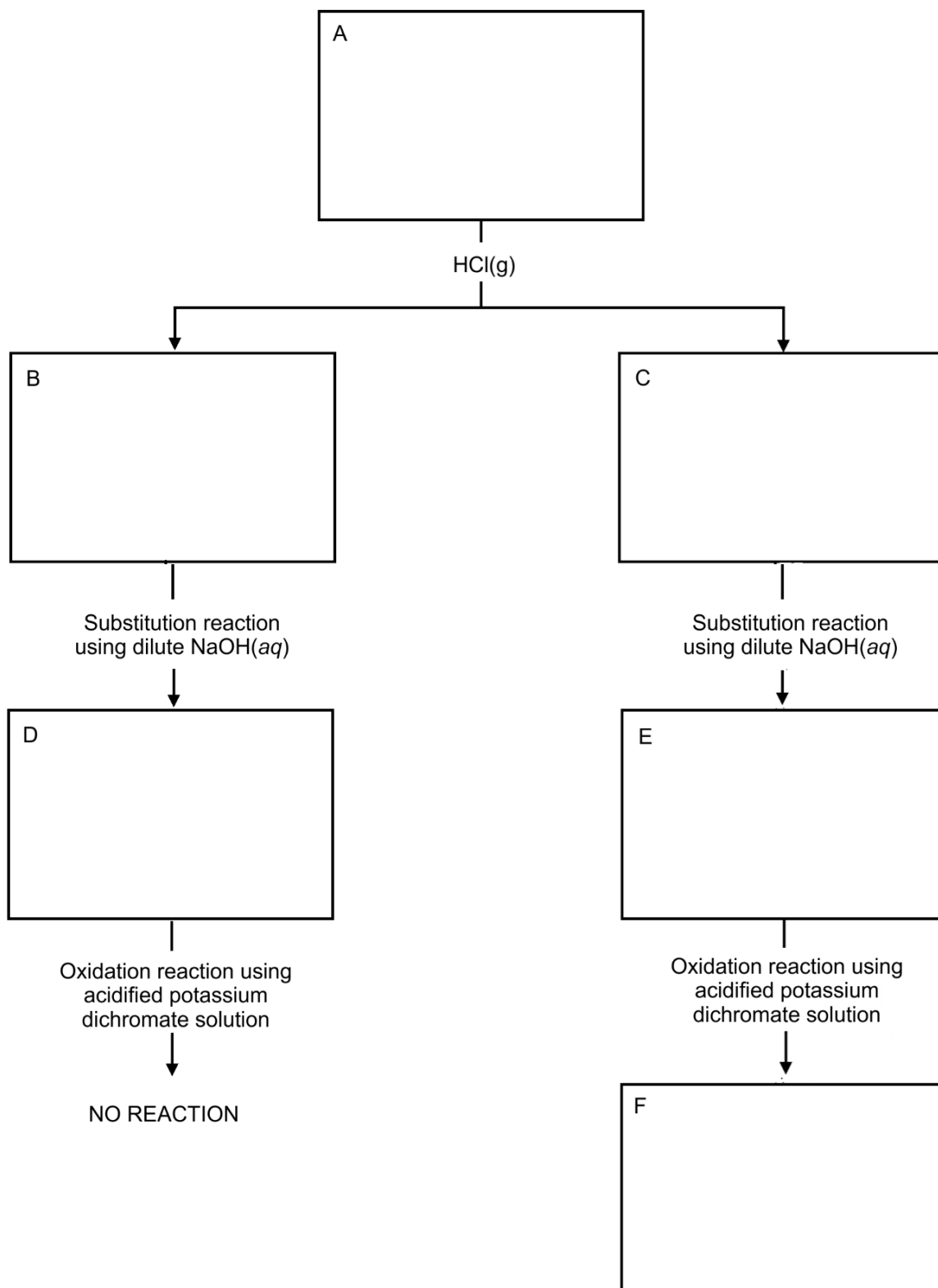
- (b) The student can use a combination of infrared, mass spectrometry and  $^{13}\text{C}$  NMR to confirm the contents of the three bottles. Evaluate the suitability of these techniques for this purpose.

6

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**Question 32** (5 marks)**Marks**

Alkene A reacts with gaseous hydrogen chloride to form two isomers, B and C, with molecular formula  $C_4H_9Cl$ . A sequence of reactions for each compound is shown below. Determine the structural formulae of the compounds A to F.



|                         |  |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|--|
|                         |  |  |  |  |  |  |  |
| <b>CANDIDATE NUMBER</b> |  |  |  |  |  |  |  |

**Question 33** (5 marks)

## Marks

Periclase is a mineral composed of magnesium oxide ( $MM\ 40.31\text{ g mol}^{-1}$ ). A  $0.4802\text{ g}$  ore sample containing periclase is treated with  $50.00\text{ mL}$  of  $0.5020\text{ mol L}^{-1}$  hydrochloric acid. The acid reacts with the magnesium oxide, but not with the other components of the ore sample. After reaction and filtration, the solution is made up to  $250.0\text{ mL}$  in a volumetric flask.  $25.00\text{ mL}$  aliquots of this solution are titrated with  $0.1034\text{ mol L}^{-1}$  sodium hydroxide ( $\text{NaOH}$ ), requiring an average of  $12.90\text{ mL}$  to reach the endpoint.

Calculate the percentage by mass of magnesium oxide in the periclase ore sample.

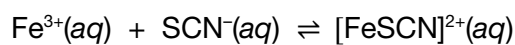
**5**

[illegible]

**Question 34** (5 marks)

## Mark

Red iron(III) thiocyanate is produced in the following equilibrium reaction.



At equilibrium at 298 K there are 0.00540 mol of  $\text{Fe}^{3+}(\text{aq})$ , 0.00823 mol of  $\text{SCN}^{-}(\text{aq})$  and 0.0631 mol of  $[\text{FeSCN}]^{2+}(\text{aq})$  in a 100.0 mL solution.

What mass of pure sodium hydroxide should be added to increase the amount of  $\text{SCN}^-(aq)$  to 0.00944 mol at equilibrium?

**5**

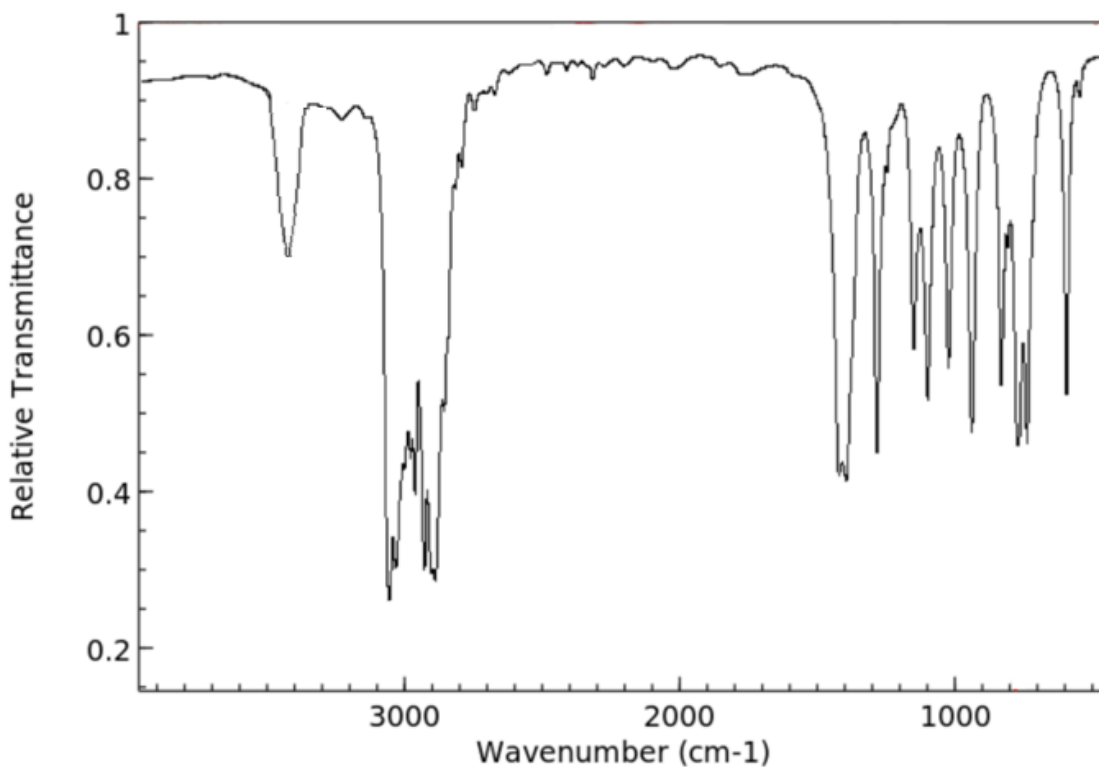
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**Question 35** (7 marks)

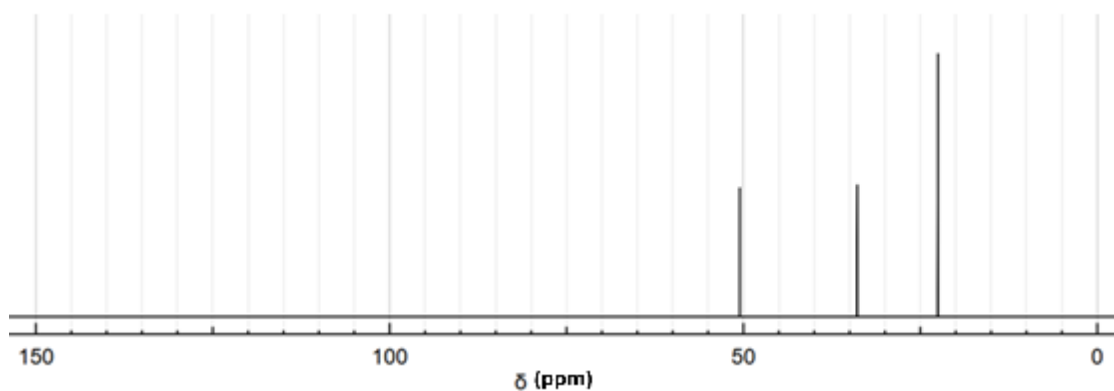
Compound Q is an organic compound which contains roughly 19% nitrogen by mass.

The infrared,  $^{13}\text{C}$  NMR and  $^1\text{H}$  NMR spectra of compound Q are shown below.

Chemical shift data for  $^1\text{H}$  NMR is provided.



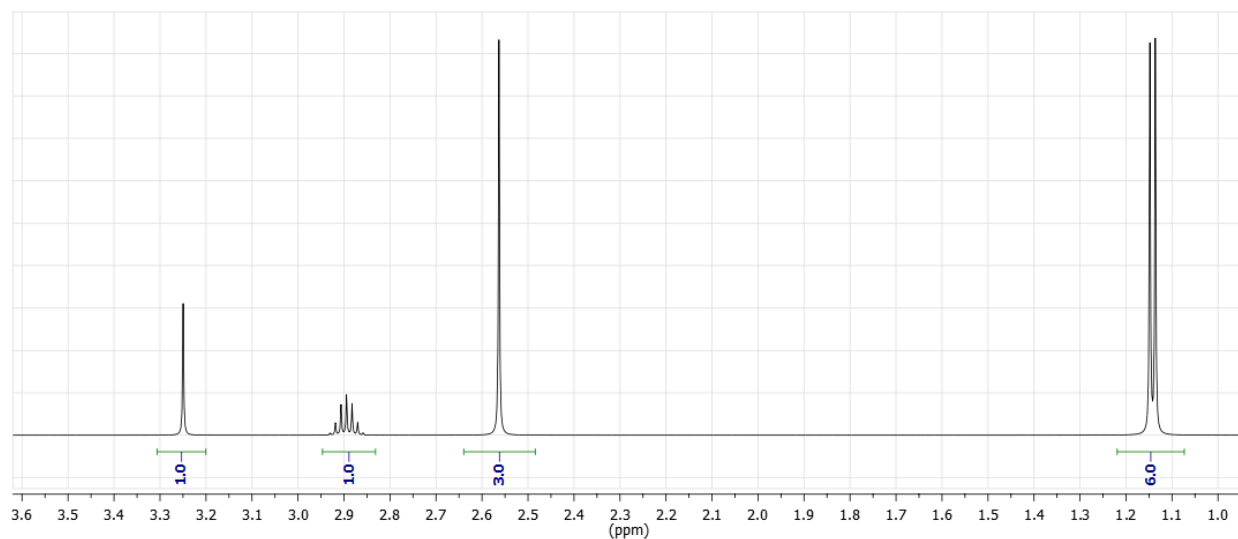
*IR spectrum of compound Q*



*$^{13}\text{C}$  NMR spectrum of compound Q*

**Question continues on next page.**

## Question 35 continued.

 $^1\text{H}$  NMR spectrum of compound Q $^1\text{H}$  NMR chemical shift data

| Type of proton                                                 | $\delta/\text{ppm}$ |
|----------------------------------------------------------------|---------------------|
| $\text{Si}(\text{CH}_3)_4$ (TMS)                               | 0                   |
| $\text{R}-\text{CH}_3$                                         | 0.9–1.0             |
| $\text{R}-\text{CH}_2-\text{R}$                                | 1.2–1.5             |
| $\text{R}-\text{CHR}_2$                                        | 1.5–2.0             |
| $\text{R}-\text{C}\equiv\text{C}-\text{H}$ (alkyne)            | 2.0–3.1             |
| $-\text{CO}-\text{CH}_2-$ (aldehydes, ketones or esters)       | 2.1–2.7             |
| $\text{R}-\text{CH}_2-\text{NH}_2$ or $\text{CHR}-\text{NH}_2$ | 2.4–3.0             |
| $\text{R}-\text{CH}_2-\text{NHR}$ or $\text{CHR}-\text{NHR}$   | 2.5–3.1             |
| $-\text{CH}_2-\text{O}-$ (alcohols, ethers or esters)          | 3.3–4.8             |
| $\text{R}-\text{OH}$                                           | 1–6                 |
| $\text{R}-\text{NH}_2$                                         | 1–5                 |
| $\text{R}-\text{NHR}$                                          | 1–5                 |
| $\text{R}_2\text{C}=\text{CHR}$ (alkene)                       | 4.5–7.0             |
| $\text{R}-\text{COONH}-\text{R}$ (amide)                       | 5–9                 |
| $\text{R}-\text{CHO}$ (aldehyde)                               | 9.4–10.0            |
| $\text{R}-\text{COOH}$                                         | 9.0–13.0            |

**Question 35 continued.**

## Marks

Draw and name compound Q and justify your choice using all information provided.

7

Compound Q

Name:

**Question 35 continued.**

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

2019 HIGHER SCHOOL CERTIFICATE  
EXAMINATION

## Chemistry

### FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mC\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{v}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant,  $N_A$  .....  $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at  $0^\circ\text{C}$  (273.15 K) ..... 22.71 L

at  $25^\circ\text{C}$  (298.15 K) ..... 24.79 L

Gas constant .....  $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at  $25^\circ\text{C}$  (298.15 K),  $K_w$  .....  $1.0 \times 10^{-14}$

Specific heat capacity of water .....  $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

### DATA SHEET

#### Solubility constants at $25^\circ\text{C}$

| Compound             | $K_{sp}$               | Compound            | $K_{sp}$               |
|----------------------|------------------------|---------------------|------------------------|
| Barium carbonate     | $2.58 \times 10^{-9}$  | Lead(II) bromide    | $6.60 \times 10^{-6}$  |
| Barium hydroxide     | $2.55 \times 10^{-4}$  | Lead(II) chloride   | $1.70 \times 10^{-5}$  |
| Barium phosphate     | $1.3 \times 10^{-29}$  | Lead(II) iodide     | $9.8 \times 10^{-9}$   |
| Barium sulfate       | $1.08 \times 10^{-10}$ | Lead(II) carbonate  | $7.40 \times 10^{-14}$ |
| Calcium carbonate    | $3.36 \times 10^{-9}$  | Lead(II) hydroxide  | $1.43 \times 10^{-15}$ |
| Calcium hydroxide    | $5.02 \times 10^{-6}$  | Lead(II) phosphate  | $8.0 \times 10^{-43}$  |
| Calcium phosphate    | $2.07 \times 10^{-29}$ | Lead(II) sulfate    | $2.53 \times 10^{-8}$  |
| Calcium sulfate      | $4.93 \times 10^{-5}$  | Magnesium carbonate | $6.82 \times 10^{-6}$  |
| Copper(II) carbonate | $1.4 \times 10^{-10}$  | Magnesium hydroxide | $5.61 \times 10^{-12}$ |
| Copper(II) hydroxide | $2.2 \times 10^{-20}$  | Magnesium phosphate | $1.04 \times 10^{-24}$ |
| Copper(II) phosphate | $1.40 \times 10^{-37}$ | Silver bromide      | $5.35 \times 10^{-13}$ |
| Iron(II) carbonate   | $3.13 \times 10^{-11}$ | Silver chloride     | $1.77 \times 10^{-10}$ |
| Iron(II) hydroxide   | $4.87 \times 10^{-17}$ | Silver carbonate    | $8.46 \times 10^{-12}$ |
| Iron(III) hydroxide  | $2.79 \times 10^{-39}$ | Silver hydroxide    | $2.0 \times 10^{-8}$   |
| Iron(III) phosphate  | $9.91 \times 10^{-16}$ | Silver iodide       | $8.52 \times 10^{-17}$ |
|                      |                        | Silver phosphate    | $8.89 \times 10^{-17}$ |
|                      |                        | Silver sulfate      | $1.20 \times 10^{-5}$  |

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

**Infrared absorption data**

| Bond              | Wavenumber/cm <sup>-1</sup> |
|-------------------|-----------------------------|
| N—H<br>(amines)   | 3300–3500                   |
| O—H<br>(alcohols) | 3230–3550<br>(broad)        |
| C—H               | 2850–3300                   |
| O—H<br>(acids)    | 2500–3000<br>(very broad)   |
| C≡N               | 2220–2260                   |
| C=O               | 1680–1750                   |
| C=C               | 1620–1680                   |
| C—O               | 1000–1300                   |
| C—C               | 750–1100                    |

**<sup>13</sup>C NMR chemical shift data**

| Type of carbon                                                                           | δ/ppm   |
|------------------------------------------------------------------------------------------|---------|
| $\begin{array}{c}   \quad   \\ -C-C- \\   \quad   \end{array}$                           | 5–40    |
| $\begin{array}{c}   \\ R-C-Cl \text{ or } Br \\   \end{array}$                           | 10–70   |
| $\begin{array}{c}   \\ R-C-C- \\    \quad   \\ O \end{array}$                            | 20–50   |
| $\begin{array}{c}   \\ R-C-N \\   \quad \diagup \quad \diagdown \end{array}$             | 25–60   |
| $\begin{array}{c}   \\ -C-O- \\   \end{array}$ alcohols,<br>ethers or<br>esters          | 50–90   |
| $\begin{array}{c} \diagup \quad \diagdown \\ C=C \\ \diagdown \quad \diagup \end{array}$ | 90–150  |
| $R-C \equiv N$                                                                           | 110–125 |
|       | 110–160 |
| $\begin{array}{c} R-C- \\    \\ O \end{array}$ esters or<br>acids                        | 160–185 |
| $\begin{array}{c} R-C- \\    \\ O \end{array}$ aldehydes<br>or ketones                   | 190–220 |

**UV absorption***(This is not a definitive list and is approximate.)*

| Chromophore | $\lambda_{\max}$ (nm) |
|-------------|-----------------------|
| C—H         | 122                   |
| C—C         | 135                   |
| C=C         | 162                   |

| Chromophore | $\lambda_{\max}$ (nm) |
|-------------|-----------------------|
| C≡C         | 173 178<br>196 222    |
| C—Cl        | 173                   |
| C—Br        | 208                   |

**Some standard potentials**

|                                         |                      |                             |         |
|-----------------------------------------|----------------------|-----------------------------|---------|
| $K^+ + e^-$                             | $\rightleftharpoons$ | $K(s)$                      | -2.94 V |
| $Ba^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Ba(s)$                     | -2.91 V |
| $Ca^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Ca(s)$                     | -2.87 V |
| $Na^+ + e^-$                            | $\rightleftharpoons$ | $Na(s)$                     | -2.71 V |
| $Mg^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Mg(s)$                     | -2.36 V |
| $Al^{3+} + 3e^-$                        | $\rightleftharpoons$ | $Al(s)$                     | -1.68 V |
| $Mn^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Mn(s)$                     | -1.18 V |
| $H_2O + e^-$                            | $\rightleftharpoons$ | $\frac{1}{2}H_2(g) + OH^-$  | -0.83 V |
| $Zn^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Zn(s)$                     | -0.76 V |
| $Fe^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Fe(s)$                     | -0.44 V |
| $Ni^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Ni(s)$                     | -0.24 V |
| $Sn^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Sn(s)$                     | -0.14 V |
| $Pb^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Pb(s)$                     | -0.13 V |
| $H^+ + e^-$                             | $\rightleftharpoons$ | $\frac{1}{2}H_2(g)$         | 0.00 V  |
| $SO_4^{2-} + 4H^+ + 2e^-$               | $\rightleftharpoons$ | $SO_2(aq) + 2H_2O$          | 0.16 V  |
| $Cu^{2+} + 2e^-$                        | $\rightleftharpoons$ | $Cu(s)$                     | 0.34 V  |
| $\frac{1}{2}O_2(g) + H_2O + 2e^-$       | $\rightleftharpoons$ | $2OH^-$                     | 0.40 V  |
| $Cu^+ + e^-$                            | $\rightleftharpoons$ | $Cu(s)$                     | 0.52 V  |
| $\frac{1}{2}I_2(s) + e^-$               | $\rightleftharpoons$ | $I^-$                       | 0.54 V  |
| $\frac{1}{2}I_2(aq) + e^-$              | $\rightleftharpoons$ | $I^-$                       | 0.62 V  |
| $Fe^{3+} + e^-$                         | $\rightleftharpoons$ | $Fe^{2+}$                   | 0.77 V  |
| $Ag^+ + e^-$                            | $\rightleftharpoons$ | $Ag(s)$                     | 0.80 V  |
| $\frac{1}{2}Br_2(l) + e^-$              | $\rightleftharpoons$ | $Br^-$                      | 1.08 V  |
| $\frac{1}{2}Br_2(aq) + e^-$             | $\rightleftharpoons$ | $Br^-$                      | 1.10 V  |
| $\frac{1}{2}O_2(g) + 2H^+ + 2e^-$       | $\rightleftharpoons$ | $H_2O$                      | 1.23 V  |
| $\frac{1}{2}Cl_2(g) + e^-$              | $\rightleftharpoons$ | $Cl^-$                      | 1.36 V  |
| $\frac{1}{2}Cr_2O_7^{2-} + 7H^+ + 3e^-$ | $\rightleftharpoons$ | $Cr^{3+} + \frac{7}{2}H_2O$ | 1.36 V  |
| $\frac{1}{2}Cl_2(aq) + e^-$             | $\rightleftharpoons$ | $Cl^-$                      | 1.40 V  |
| $MnO_4^- + 8H^+ + 5e^-$                 | $\rightleftharpoons$ | $Mn^{2+} + 4H_2O$           | 1.51 V  |
| $\frac{1}{2}F_2(g) + e^-$               | $\rightleftharpoons$ | $F^-$                       | 2.89 V  |

## PERIODIC TABLE OF THE ELEMENTS

| PERIODIC TABLE OF THE ELEMENTS |                                |                                                           |                                |                               |                                 |                                |                                |                              |                                |                             |                              |                               |                                |                                |                                |                               |                              | 2<br>He<br>4.003<br>Helium |
|--------------------------------|--------------------------------|-----------------------------------------------------------|--------------------------------|-------------------------------|---------------------------------|--------------------------------|--------------------------------|------------------------------|--------------------------------|-----------------------------|------------------------------|-------------------------------|--------------------------------|--------------------------------|--------------------------------|-------------------------------|------------------------------|----------------------------|
| 1<br>H<br>1.008<br>Hydrogen    | KEY                            |                                                           |                                |                               |                                 |                                |                                |                              |                                |                             |                              |                               |                                |                                |                                |                               |                              |                            |
| 3<br>Li<br>6.941<br>Lithium    | 4<br>Be<br>9.012<br>Beryllium  | Atomic Number<br>Symbol<br>Standard Atomic Weight<br>Name |                                |                               |                                 |                                |                                |                              |                                |                             |                              | 79<br>Au<br>197.0<br>Gold     | 5<br>B<br>10.81<br>Boron       | 6<br>C<br>12.01<br>Carbon      | 7<br>N<br>14.01<br>Nitrogen    | 8<br>O<br>16.00<br>Oxygen     | 9<br>F<br>19.00<br>Fluorine  | 10<br>Ne<br>20.18<br>Neon  |
| 11<br>Na<br>22.99<br>Sodium    | 12<br>Mg<br>24.31<br>Magnesium |                                                           |                                |                               |                                 |                                |                                |                              |                                |                             |                              | 13<br>Al<br>26.98<br>Aluminum | 14<br>Si<br>28.09<br>Silicon   | 15<br>P<br>30.97<br>Phosphorus | 16<br>S<br>32.07<br>Sulfur     | 17<br>Cl<br>35.45<br>Chlorine | 18<br>Ar<br>39.95<br>Argon   |                            |
| 19<br>K<br>39.10<br>Potassium  | 20<br>Ca<br>40.08<br>Calcium   | 21<br>Sc<br>44.96<br>Scandium                             | 22<br>Ti<br>47.87<br>Titanium  | 23<br>V<br>50.94<br>Vanadium  | 24<br>Cr<br>52.00<br>Chromium   | 25<br>Mn<br>54.94<br>Manganese | 26<br>Fe<br>55.85<br>Iron      | 27<br>Co<br>58.93<br>Cobalt  | 28<br>Ni<br>58.69<br>Nickel    | 29<br>Cu<br>63.55<br>Copper | 30<br>Zn<br>65.38<br>Zinc    | 31<br>Ga<br>69.72<br>Gallium  | 32<br>Ge<br>72.64<br>Germanium | 33<br>As<br>74.92<br>Arsenic   | 34<br>Se<br>78.96<br>Selenium  | 35<br>Br<br>79.90<br>Bromine  | 36<br>Kr<br>83.80<br>Krypton |                            |
| 37<br>Rb<br>85.47<br>Rubidium  | 38<br>Sr<br>87.61<br>Strontium | 39<br>Y<br>88.91<br>Yttrium                               | 40<br>Zr<br>91.22<br>Zirconium | 41<br>Nb<br>92.91<br>Niobium  | 42<br>Mo<br>95.96<br>Molybdenum | 43<br>Tc<br>Technetium         | 44<br>Ru<br>101.1<br>Ruthenium | 45<br>Rh<br>102.9<br>Rhodium | 46<br>Pd<br>106.4<br>Palladium | 47<br>Ag<br>107.9<br>Silver | 48<br>Cd<br>112.4<br>Cadmium | 49<br>In<br>114.8<br>Indium   | 50<br>Sn<br>118.7<br>Tin       | 51<br>Sb<br>121.8<br>Antimony  | 52<br>Te<br>127.6<br>Tellurium | 53<br>I<br>126.9<br>Iodine    | 54<br>Xe<br>131.3<br>Xenon   |                            |
| 55<br>Cs<br>132.9<br>Caesium   | 56<br>Ba<br>137.3<br>Barium    | 57–71<br>Lanthanoids                                      | 72<br>Hf<br>178.5<br>Hafnium   | 73<br>Ta<br>180.9<br>Tantalum | 74<br>W<br>183.9<br>Tungsten    | 75<br>Re<br>186.2<br>Rhenium   | 76<br>Os<br>190.2<br>Osmium    | 77<br>Ir<br>192.2<br>Iridium | 78<br>Pt<br>195.1<br>Platinum  | 79<br>Au<br>197.0<br>Gold   | 80<br>Hg<br>200.6<br>Mercury | 81<br>Tl<br>204.4<br>Thallium | 82<br>Pb<br>207.2<br>Lead      | 83<br>Bi<br>209.0<br>Bismuth   | 84<br>Po<br>Polonium           | 85<br>At<br>Astatine          | 86<br>Rn<br>Radon            |                            |
| 87<br>Fr<br>Francium           | 88<br>Ra<br>Radium             | 89–103<br>Actinoids                                       | 104<br>Rf<br>Rutherfordium     | 105<br>Db<br>Dubnium          | 106<br>Sg<br>Seaborgium         | 107<br>Bh<br>Bohrium           | 108<br>Hs<br>Hassium           | 109<br>Mt<br>Meitnerium      | 110<br>Ds<br>Darmstadtium      | 111<br>Rg<br>Roentgenium    | 112<br>Cn<br>Copernicium     | 113<br>Nh<br>Nihonium         | 114<br>Fl<br>Flerovium         | 115<br>Mc<br>Moscovium         | 116<br>Lv<br>Livermorium       | 117<br>Ts<br>Tennessine       | 118<br>Og<br>Oganesson       |                            |

## Lanthanoids

|                                |                             |                                   |                                |                        |                               |                               |                                 |                              |                                 |                              |                             |                              |                                |                               |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|
| 57<br>La<br>138.9<br>Lanthanum | 58<br>Ce<br>140.1<br>Cerium | 59<br>Pr<br>140.9<br>Praseodymium | 60<br>Nd<br>144.2<br>Neodymium | 61<br>Pm<br>Promethium | 62<br>Sm<br>150.4<br>Samarium | 63<br>Eu<br>152.0<br>Europium | 64<br>Gd<br>157.3<br>Gadolinium | 65<br>Tb<br>158.9<br>Terbium | 66<br>Dy<br>162.5<br>Dysprosium | 67<br>Ho<br>164.9<br>Holmium | 68<br>Er<br>167.3<br>Erbium | 69<br>Tm<br>168.9<br>Thulium | 70<br>Yb<br>173.1<br>Ytterbium | 71<br>Lu<br>175.0<br>Lutetium |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|

## Actinoids

|                      |                              |                                   |                             |                       |                       |                       |                    |                       |                         |                         |                      |                          |                       |                         |
|----------------------|------------------------------|-----------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|--------------------|-----------------------|-------------------------|-------------------------|----------------------|--------------------------|-----------------------|-------------------------|
| 89<br>Ac<br>Actinium | 90<br>Th<br>232.0<br>Thorium | 91<br>Pa<br>231.0<br>Protactinium | 92<br>U<br>238.0<br>Uranium | 93<br>Np<br>Neptunium | 94<br>Pu<br>Plutonium | 95<br>Am<br>Americium | 96<br>Cm<br>Curium | 97<br>Bk<br>Berkelium | 98<br>Cf<br>Californium | 99<br>Es<br>Einsteinium | 100<br>Fm<br>Fermium | 101<br>Md<br>Mendelevium | 102<br>No<br>Nobelium | 103<br>Lr<br>Lawrencium |
|----------------------|------------------------------|-----------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|--------------------|-----------------------|-------------------------|-------------------------|----------------------|--------------------------|-----------------------|-------------------------|

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.



TRIAL EXAMINATION 2025

# FORM VI CHEMISTRY

**CRIB**

## General Instructions

- Reading Time: 5 minutes
- Working Time: 3 hours
- Write using black pen

## Structure of Paper & Instructions

- Section 1: Multiple-choice (20 marks).  
Spend 35 minutes on this section.
- Section 2: Written answer (80 marks).  
Spend 2 hours 25 minutes on this section.

## Exam Instructions

- Remove the centre staple and hand in all parts of the paper in a neat bundle.
- Show all relevant working.
- **Write your candidate number in the space provided on pages 1, 11, 17, 23 and 27.**

**Total marks: 100**

**Friday 8 August**  
**12:00 pm**

## Checklist

Each boy should have the following:

- ☐ 1 question paper
- ☐ 1 multiple-choice answer sheet
- ☐ 1 data sheet

JL Senczyszyn

Examiners: EJS, JAC, JLS, TW

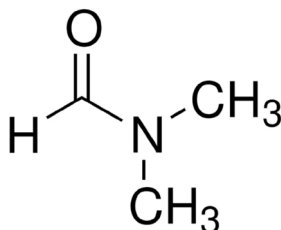
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## SECTION I: MULTIPLE CHOICE QUESTIONS

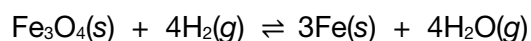
20 marks

Attempt ALL Questions  
Use the Multiple-Choice Answer Sheet.

- 1 What is the IUPAC name for the structure shown?



- (A) *N*-dimethylmethanal  
(B) *N*-ethylmethanamide  
(C) *N,N*-dimethylmethanamine  
(D) *N,N*-dimethylmethanamide
- 2 Which of the following compounds is produced in the reaction between hydrochloric acid and calcium carbonate but **not** in the reaction between hydrochloric acid and calcium hydroxide?
- (A) calcium chloride  
(B) carbon dioxide  
(C) hydrogen  
(D) water
- 3 What is the equilibrium expression for the equilibrium system shown?



- (A)  $K_{eq} = \frac{1}{[\text{H}_2]^4}$   
(B)  $K_{eq} = \frac{[\text{H}_2\text{O}]}{[\text{H}_2]}$   
(C)  $K_{eq} = \frac{[\text{H}_2\text{O}]^4}{[\text{H}_2]^4}$   
(D)  $K_{eq} = \frac{[\text{Fe}]^3[\text{H}_2\text{O}]^4}{[\text{Fe}_3\text{O}_4][\text{H}_2]^4}$

4 Which of the following substances is an Arrhenius base?

- (A)  $\text{Li}_2\text{CO}_3$
- (B)  $\text{NH}_3$
- (C)  $\text{NaHCO}_3$
- (D)  $\text{Sr}(\text{OH})_2$

5 Which compound listed below will react in the presence of concentrated sulfuric acid with the product of its own oxidation to produce an ester?

- (A) propan-1-ol
- (B) propan-2-ol
- (C) propanal
- (D) propanoic acid

6 Which compound will produce a peak at 210 ppm in  $^{13}\text{C}$  NMR spectroscopy?

- (A) pentan-1-ol
- (B) pentan-1-amine
- (C) pentanoic acid
- (D) pentanal

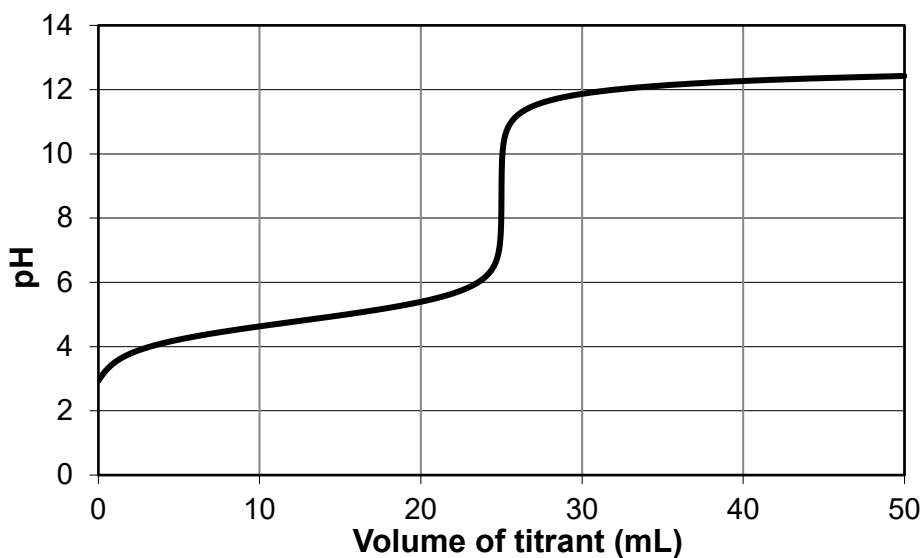
7 Consider the equilibrium system shown for the production of nitrosyl chloride (NOCl).



Which of the following conditions will maximise the equilibrium yield of nitrosyl chloride?

|     | <i>Pressure</i> | <i>Temperature</i> |
|-----|-----------------|--------------------|
| (A) | high            | high               |
| (B) | high            | low                |
| (C) | low             | high               |
| (D) | low             | low                |

Use the information below to answer questions 8 and 9.



8 Which of the following titrations would result in this titration curve?

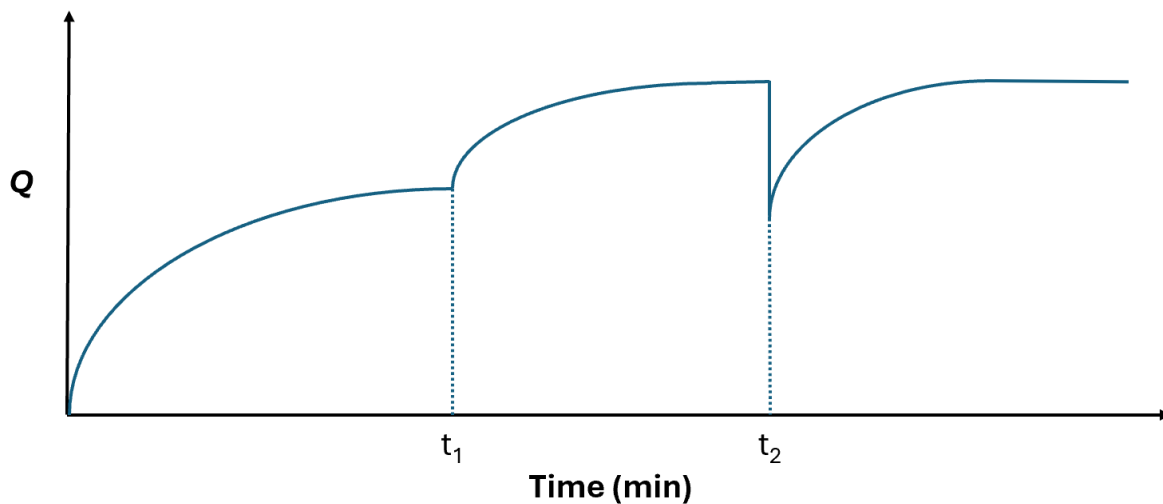
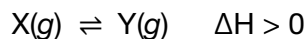
- (A)  $0.1 \text{ mol L}^{-1} \text{ C}_2\text{H}_5\text{COOH}$  with  $0.1 \text{ mol L}^{-1} \text{ NH}_3$
- (B)  $0.1 \text{ mol L}^{-1} \text{ C}_2\text{H}_5\text{COOH}$  with  $0.1 \text{ mol L}^{-1} \text{ NaOH}$
- (C)  $0.1 \text{ mol L}^{-1} \text{ HCl}$  with  $0.1 \text{ mol L}^{-1} \text{ NaOH}$
- (D)  $0.1 \text{ mol L}^{-1} \text{ HCl}$  with  $0.1 \text{ mol L}^{-1} \text{ NH}_3$

9 Which of the following indicators is most suitable for determining the endpoint of the titration above?

|     | Indicator         | Colour change pH range |
|-----|-------------------|------------------------|
| (A) | Methyl orange     | 3.1 – 4.4              |
| (B) | Bromothymol blue  | 6.0 – 7.6              |
| (C) | Phenolphthalein   | 8.2 – 10.0             |
| (D) | Alizarin yellow R | 10.1 – 12.0            |

- 10 An aqueous solution containing which substance is most likely to have a pH of 9?
- (A) methanol
  - (B) propanoic acid
  - (C) ethanamine
  - (D) butanal
- 11 In proton NMR, what do the relative peak areas indicate about a pure substance?
- (A) The number of protons in the molecule.
  - (B) The ratio of protons in different environments.
  - (C) The proximity of the proton to an electronegative element.
  - (D) The molecular weight of the molecule.
- 12 Infrared spectroscopy of a pure compound produces a broad peak centered at  $3415\text{ cm}^{-1}$ . Which series of organic compounds is the unknown compound likely to belong to?
- (A) alcohol
  - (B) carboxylic acid
  - (C) aldehyde
  - (D) alkene
- 13 Which of the following mixtures of aqueous species will form a buffer solution when both species have a concentration of approximately  $0.1\text{ mol L}^{-1}$ .
- (A)  $\text{HCl} / \text{Cl}^-$
  - (B)  $\text{HCOOH} / \text{HCOO}^-$
  - (C)  $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$
  - (D)  $\text{H}_2\text{O} / \text{OH}^-$

14 The graph shows changes in  $Q$  over time for the equilibrium system shown.



Which of the following could be the changes made at  $t_1$  and  $t_2$ ?

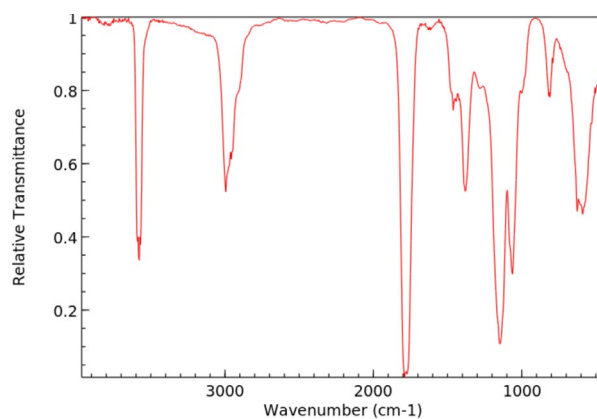
|     | <i>Change made at <math>t_1</math></i> | <i>Change made at <math>t_2</math></i> |
|-----|----------------------------------------|----------------------------------------|
| (A) | Decrease in temperature                | Addition of X                          |
| (B) | Decrease in temperature                | Removal of X                           |
| (C) | Increase in temperature                | Addition of Y                          |
| (D) | Increase in temperature                | Removal of Y                           |

15 Which set of peaks would be likely found in the mass spectrum of 2,2-dimethylpropanal?

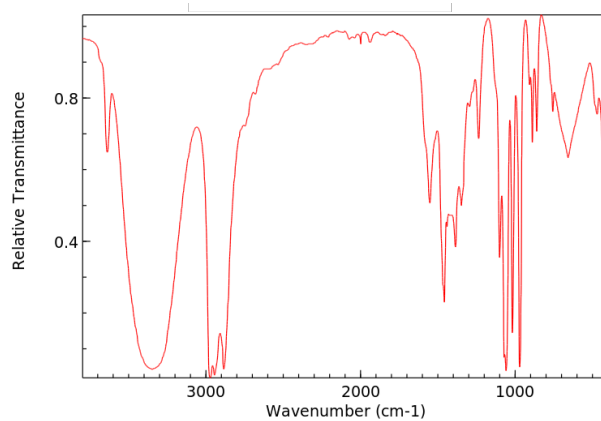
- (A)  $m/z = 14$ ,  $m/z = 28$ ,  $m/z = 57$ ,  $m/z = 71$  and  $m/z = 85$
- (B)  $m/z = 14$ ,  $m/z = 29$ ,  $m/z = 58$ ,  $m/z = 73$  and  $m/z = 87$
- (C)  $m/z = 15$ ,  $m/z = 29$ ,  $m/z = 57$ ,  $m/z = 71$  and  $m/z = 86$
- (D)  $m/z = 15$ ,  $m/z = 28$ ,  $m/z = 58$ ,  $m/z = 73$  and  $m/z = 86$

16 Which infrared spectrum was most likely produced from a pure sample of propanone?

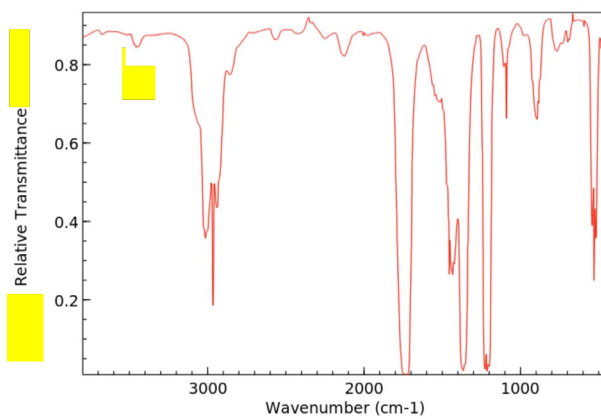
(A)



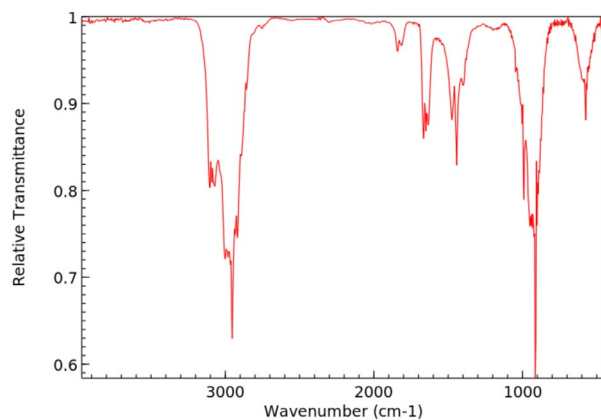
(B)



(C)



(D)



- 17** In an experiment to determine the  $K_{sp}$  value of barium fluoride, five 1.0 mL samples containing barium ions were prepared and each was added to a 1.0 mL solution containing fluoride ions. The results are given below.

| Sample | Barium ion solution                          |             | Fluoride ion solution                        |             | Observation             |
|--------|----------------------------------------------|-------------|----------------------------------------------|-------------|-------------------------|
|        | Initial concentration (mol L <sup>-1</sup> ) | Volume (mL) | Initial concentration (mol L <sup>-1</sup> ) | Volume (mL) |                         |
| 1      | 0.010                                        | 1.0         | 0.026                                        | 1.0         | No precipitate          |
| 2      | 0.010                                        | 1.0         | 0.028                                        | 1.0         | No precipitate          |
| 3      | 0.010                                        | 1.0         | 0.030                                        | 1.0         | Faint white precipitate |
| 4      | 0.010                                        | 1.0         | 0.032                                        | 1.0         | White precipitate       |

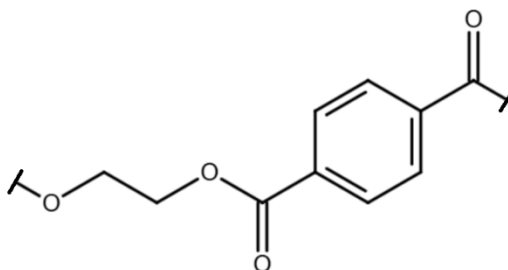
Which of the following could be the  $K_{sp}$  value of barium fluoride?

- (A)  $1.0 \times 10^{-6}$
- (B)  $1.2 \times 10^{-6}$
- (C)  $8.8 \times 10^{-6}$
- (D)  $7.2 \times 10^{-5}$
- 18** When ethene is reacted completely with hydrogen bromide, 7.27 L of the product is obtained at 25.0°C and 120 kPa. What is the mass of ethene that was reacted?
- (A) 7.27 g
- (B) 9.87 g
- (C) 79.7 g
- (D) 118 g

**19** A salt has a  $K_{sp}$  value of  $4.3 \times 10^{-10}$  and a molar solubility of  $2.0 \times 10^{-3} \text{ mol L}^{-1}$ . Which of the following represents the formula of this salt?

- (A) XY
- (B)  $\text{XY}_2$
- (C)  $\text{XY}_3$
- (D)  $\text{X}_2\text{Y}_3$

**20** 1.242 kg of a single polymer with the repeating unit shown below was obtained after a polymerisation reaction.



What mass of water is released during this polymerisation reaction?

- (A) 116 g
- (B) 233 g
- (C) 656 g
- (D) 803 g

## SECTION II: WRITTEN ANSWER QUESTIONS

80 marks

Attempt ALL Questions

|                  |  |  |  |  |  |  |  |
|------------------|--|--|--|--|--|--|--|
|                  |  |  |  |  |  |  |  |
| CANDIDATE NUMBER |  |  |  |  |  |  |  |

### Question 21 (2 marks)

Marks

Using balanced chemical equations, show that the  $\text{H}_2\text{PO}_4^-$  ion can react with both strong acids and strong bases.

2



Accepted  $\text{H}_3\text{O}^+$  or  $\text{H}^+$ , any strong acid such as  $\text{HCl}$ , any reasonable metal hydroxide. Equations needed to clearly show the correct gain/loss of  $\text{H}^+$ , and have correct balancing.

### Question 22 (3 marks)

Calculate the pH of  $0.20 \text{ mol L}^{-1}$  hydrazoic acid ( $\text{HN}_3$ ,  $K_a 1.9 \times 10^{-5}$ ).

3

|   | $\text{HN}_3(aq) \rightleftharpoons$ | $\text{H}^+(aq)$ | $+ \text{N}_3^-(aq)$ |
|---|--------------------------------------|------------------|----------------------|
| I | 0.20                                 | 0                | 0                    |
| C | $-x$                                 | $+x$             | $+x$                 |
| E | $0.20 - x$                           | $x$              | $x$                  |

$$K_a = \frac{[\text{H}^+(aq)][\text{N}_3^-(aq)]}{[\text{HN}_3(aq)]}$$

Assuming that  $x \ll 0.20$ ,

$$\frac{x^2}{0.20} = 1.9 \times 10^{-5}$$

$$x^2 = 3.8 \times 10^{-6}$$

$$x = 1.9 \times 10^{-3} \text{ mol L}^{-1}$$

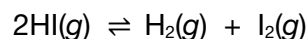
$$\text{pH} = -\log_{10}(1.9 \times 10^{-3}) = 2.71$$

(Same result if no assumption made)

|                                   |         |
|-----------------------------------|---------|
| Correct answer with working       | 3 marks |
| Overall method correct with error | 2 marks |
| Some relevant information         | 1 mark  |

**Question 23** (6 marks)**Marks**

Hydrogen iodide decomposes when heated to hydrogen and iodine as shown below.



- (a) Give the equilibrium expression for this reaction.

**1**

$$K_{eq} = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} \quad (\text{Must include } K_{eq})$$

- (b) In an experiment, 1.75 mol of hydrogen iodide was placed in a 1.00 L vessel at a constant temperature. At equilibrium, the concentration of iodine gas was  $0.168 \text{ mol L}^{-1}$ .

What is the  $K_{eq}$  value at this temperature?

**3**

|   | $2\text{HI(g)} \rightleftharpoons$ | $\text{H}_2\text{(g)} +$ | $\text{I}_2\text{(g)}$ |
|---|------------------------------------|--------------------------|------------------------|
| I | 1.75 M                             | 0                        | 0                      |
| C | $-(2 \times 0.168) \text{ M}$      | +0.168                   | +0.168                 |
| E | 1.414 M                            | 0.168 M                  | 0.168 M                |

$$K_{eq} = \frac{[\text{H}_2][\text{I}_2]}{[\text{HI}]^2} = \frac{[0.168][0.168]}{[1.414]^2} = 0.0141$$

|                                               |         |
|-----------------------------------------------|---------|
| Correct answer to 3 significant figures       | 3 marks |
| A value for $K_{eq}$ calculated with an error | 2 marks |
| Some relevant information                     | 1 mark  |

- (c) Explain what would happen to the value of  $K_{eq}$  if a catalyst was added at constant temperature.

**2**

1-  $K_{eq}$  will be the same.

1- catalyst increases the reaction rate of both the forward and reverse reaction **by the same amount** so the forward rate will be the same as the reverse rate and therefore  $K_{eq}$  remains the same

OR

$[\text{H}_2]$ ,  $[\text{I}_2]$ ,  $[\text{HI}]$  are not affected by addition of catalyst therefore  $K_{eq}$  remains the same.

Did not accept explanation by negative e.g. "only temperature changes  $K_{eq}$ ".

**Question 24** (4 marks)**Marks**

100.0 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide is mixed with 100.0 mL of 1.00 mol L<sup>-1</sup> nitric acid in a cardboard cup. The initial temperature of each solution is 24.30 °C and the resulting solution has a temperature of 31.15 °C.

You can assume that there is no volume change upon reaction, and that the resulting solution has a density of 1.04 g mL<sup>-1</sup> and a specific heat capacity of 4.03 J K<sup>-1</sup> g<sup>-1</sup>.

Calculate the enthalpy of neutralisation for this reaction.

**4**

$$m(\text{solution after reaction}) = 200.0 \text{ mL} \times 1.04 \text{ g mL}^{-1} = 208 \text{ g}$$

$$\Delta T = 31.15 - 24.30 = 6.85 \text{ K}$$

$$q = mc\Delta T = 208 \text{ g} \times 4.03 \text{ J K}^{-1} \text{ g}^{-1} \times 6.85 \text{ K} = 5740 \text{ J} = 5.74 \text{ kJ}$$

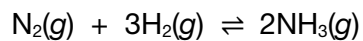
$$n(\text{H}_2\text{O}) = n(\text{HNO}_3) = n(\text{NaOH}) = 1.00 \text{ mol L}^{-1} \times 0.100 \text{ L} = 0.100 \text{ mol}$$

$$\Delta H = -5.74 \text{ kJ} / 0.100 \text{ mol} = -57.4 \text{ kJ mol}^{-1} \text{ (to 3 s.f.)}$$

|                                                                                               |         |
|-----------------------------------------------------------------------------------------------|---------|
| Correct answer with working                                                                   | 4 marks |
| Minor mistake(s) e.g. not using the given specific heat capacity, not using the density data. | 3 marks |
| At least two of the major steps of the calculation included (moles/q/ $\Delta H$ )            | 2 marks |
| Some relevant information                                                                     | 1 mark  |

**Question 25** (10 marks)**Marks**

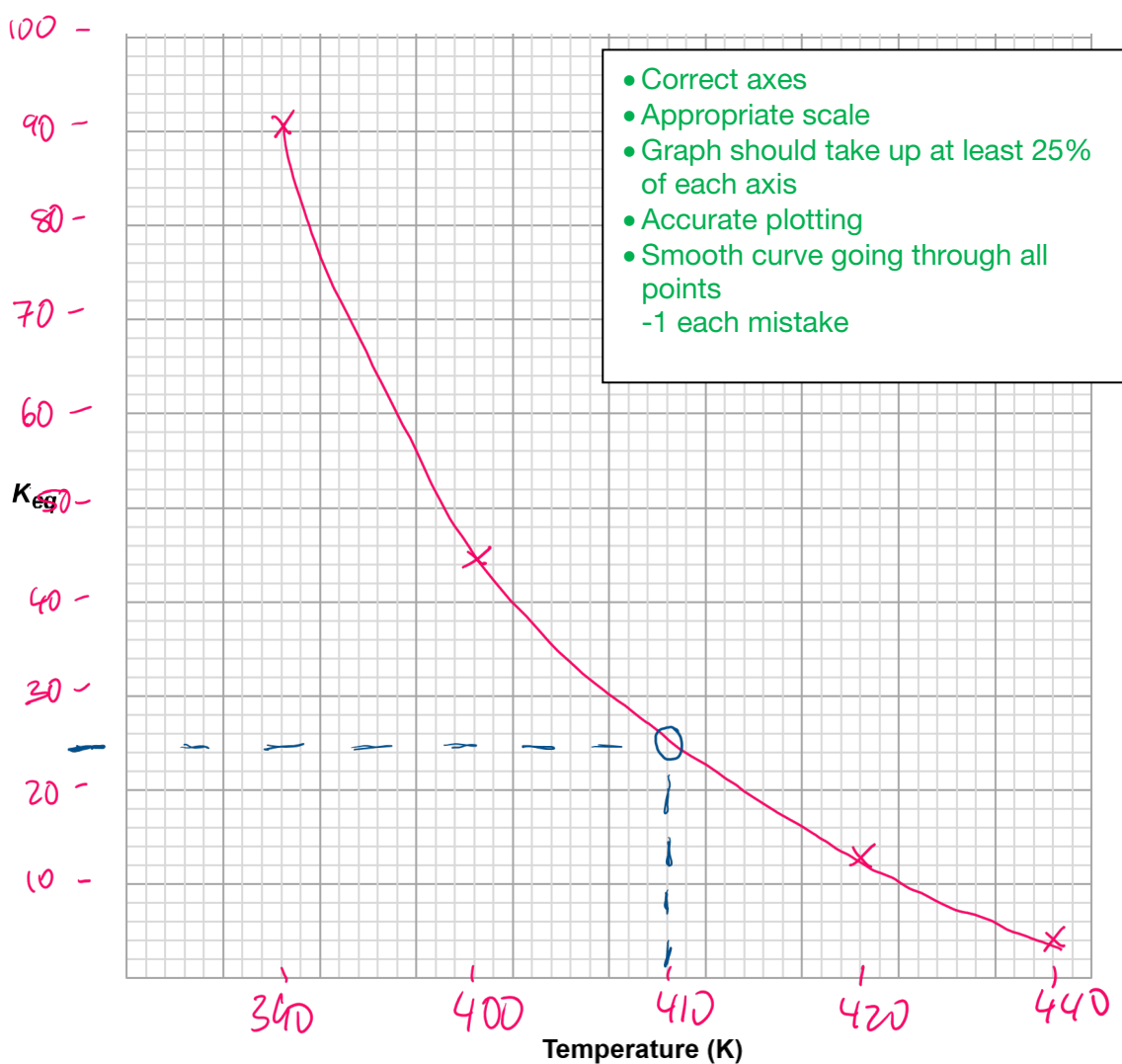
The synthesis of ammonia from its elements is an equilibrium process.



The table shows the value of  $K_{\text{eq}}$  for this reaction at a range of different temperatures.

| Temperature (K) | $K_{\text{eq}}$ |
|-----------------|-----------------|
| 390             | 90              |
| 400             | 44              |
| 420             | 12              |
| 430             | 6.4             |
| 440             | 3.6             |

(a) Plot this data, including a curved line of best fit.

**3**

**Question 25 continued.****Marks**

- (b) Using the graph, estimate
- $K_{eq}$
- at 410 K.

**1***Correct value taken from their graph (generally  $K_{eq} = 22-27$ )*

- (c) A system at 410 K is found to have
- $[N_2] = 0.24 \text{ mol L}^{-1}$
- ,
- $[H_2] = 1.1 \text{ mol L}^{-1}$
- and
- $[NH_3] = 1.6 \text{ mol L}^{-1}$
- .

Show that this system is not at equilibrium and use  $Q$  and  $K_{eq}$  to explain how the system will shift to reach equilibrium.

**3**

$$Q = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{[1.6]^2}{[0.24][1.1]^3} = 8$$

$Q < K_{eq}$ , so system is not at equilibrium.

$Q$  must increase so system will favour forward reaction, increasing  $[NH_3]$  and decreasing  $[N_2]$  and  $[H_2]$  until  $Q = K_{eq}$ .

|                                                                                     |         |
|-------------------------------------------------------------------------------------|---------|
| Correct calculation, explains shift in terms of concentration, $Q$ , and $K_{eq}$ . | 3 marks |
| Missing one of the above                                                            | 2 marks |
| Some relevant information                                                           | 1 mark  |

- (d) Is the production of ammonia exothermic or endothermic? Justify your answer using Le Châtelier's Principle and the data provided.

**3**

*Forward reaction is exothermic.*

*As seen in graph, as temperature increases  $K_{eq}$  decreases. This means as temp increases  $[N_2]$  and  $[H_2]$  increases and  $[NH_3]$  decreases and the reverse reaction is favoured.*

*According LCP, higher temperature will favour the endothermic direction, so the reverse reaction is endothermic, and the forward direction is exothermic.*

|                                                                                                                                      |         |
|--------------------------------------------------------------------------------------------------------------------------------------|---------|
| Must refer to data/graph, related LCP specifically to the scenario, explicitly conclude that the production of ammonia is exothermic | 3 marks |
| Missing part of the above                                                                                                            | 2 marks |
| Some relevant information                                                                                                            | 1 mark  |

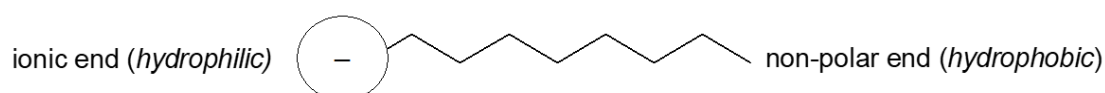
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| <b>CANDIDATE NUMBER</b> |  |  |  |  |  |  |  |

**Question 26** (5 marks)**Marks**

Soap can be used as a cleaning agent due to its ability to stabilise emulsions.

(a) Draw a labelled diagram to show the structure of a soap molecule.

**2**

| <i>Criteria</i>                                                                                                                                                              | <i>Marks</i> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Provides:<br><br>A correct diagram with a negatively charged head and a hydrocarbon tail<br>Correct labels identifying polar/hydrophilic head and non-polar/hydrophobic tail | <b>2</b>     |
| Provides one of the above                                                                                                                                                    | <b>1</b>     |

(b) Explain how soap can act as a cleaning agent.

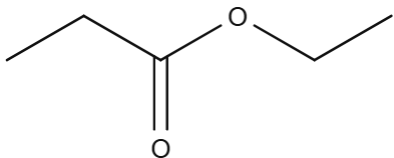
**3**

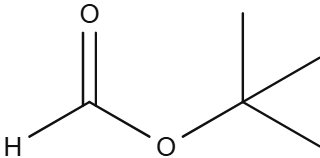
| <i>Criteria</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <i>Marks</i> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Provides a thorough explanation in terms of intermolecular forces and the formation of an emulsion in final step:<br><br>Must include: <ul style="list-style-type: none"> <li>The non-polar end of the soap ions bond to the grease via <b>dispersion forces AND</b> the ionic end bonds to water molecules via <b>H-bonding/ion-dipole</b>.</li> <li>Formation of a <b>micelle</b></li> <li>Micelles suspended in water / washed away with water – refer to cleaning action</li> <li>Must have some statement regarding keeping micelles separate</li> <li>Must refer to cleaning action – being washed away</li> </ul> | <b>3</b>     |
| Provides a sound explanation<br><br>Must have at least one intermolecular force identified or simply stated 'substance' instead of grease/oil                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>2</b>     |
| Provides some relevant information<br><br>Any correct intermolecular force<br>Identifies micelles are formed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>1</b>     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |

**Question 27** (4 marks)

Two esters, A and B, have molecular formula  $C_5H_{10}O_2$ .

Proton NMR spectroscopy was performed on both compounds, and the following data was obtained.

| <b>Ester A</b>                                                                    |                             |                           |                               |
|-----------------------------------------------------------------------------------|-----------------------------|---------------------------|-------------------------------|
|  | <i>Chemical shift (ppm)</i> | <i>Relative peak area</i> | <i>Peak splitting pattern</i> |
|                                                                                   | 1.05                        | 3                         | triplet                       |
|                                                                                   | 1.16                        | 3                         | triplet                       |
|                                                                                   | 2.26                        | 2                         | quartet                       |
|                                                                                   | 4.11                        | 2                         | quartet                       |

| <b>Ester B</b>                                                                      |                             |                           |                               |
|-------------------------------------------------------------------------------------|-----------------------------|---------------------------|-------------------------------|
|  | <i>Chemical shift (ppm)</i> | <i>Relative peak area</i> | <i>Peak splitting pattern</i> |
|                                                                                     | 1.42                        | 9                         | singlet                       |
|                                                                                     | 9.60                        | 1                         | singlet                       |

Question continues on next page.

**Question 27 continued.****Marks**

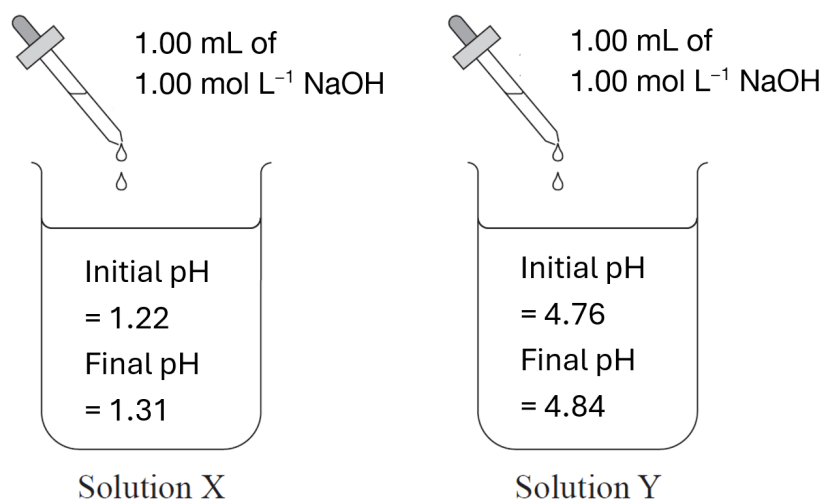
State **two** differences between the proton NMR data and relate these to the structure of the two esters.

**4**

| <b>Criteria</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Marks</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <p>States <b>two differences</b> and relates these to the structure of the two esters.</p> <p>Must have explicit link to data and structures i.e. Ester A shows.....<br/>Ester B shows.....</p> <p>Difference 1: Number of peaks<br/>Ester A produces 4 peaks ester B produces 2 peaks</p> <p>Difference 2: Different splitting<br/>Ester A produces 2 triplets and 2 quartets, while ester B produces 2 singlets<br/>Ester A must contain CH<sub>2</sub> next to CH<sub>3</sub>, whereas ester B contains two (sets of) isolated protons</p> | <b>4</b>     |
| <p>States two differences and some link to the structure</p> <p>Must have explicit links to data and one of the structures<br/>General links to both structures eg 'A' has triplets and quartets because neighbouring carbons have hydrogen but B has singlets as no neighbouring carbon has hydrogen<br/>General statement – more/less peaks</p>                                                                                                                                                                                             | <b>3</b>     |
| <p>States two differences from data with no explicit links OR states 1 difference with general link to structure</p>                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>2</b>     |
| <p>Provides some relevant information:</p> <p>States one/two differences</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>1</b>     |

**Question 28** (7 marks)**Marks**

The pH of two 100 mL solutions, X and Y, were measured before and after 1.00 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide solution was added to each.



One of these solutions contains a solution of a weak acid and its conjugate base, both at a concentration of 0.10 mol L<sup>-1</sup>; the other contains a 0.060 mol L<sup>-1</sup> solution of hydrochloric acid.

- (a) By performing a relevant calculation, show that Solution X is the 0.060 mol L<sup>-1</sup> solution of hydrochloric acid.

**2**

Any relevant calculation:

$$\text{pH} = -\log_{10}[\text{H}_3\text{O}^+] = -\log_{10}(0.0600) = 1.22$$

This is the same as the **initial** pH of solution X, so solution X is the 0.060 mol L<sup>-1</sup> solution of hydrochloric acid.

OR

$$[\text{HCl}] = [\text{H}_3\text{O}^+] = 10^{-1.22} = 0.060 \text{ mol L}^{-1}$$

The initial pH is 1.22, so the concentration of HCl must be 0.060 mol L<sup>-1</sup>

OR

$$n(\text{HCl}) = 0.006 \text{ mol} \quad n(\text{NaOH}) = 0.001 \text{ mol}$$

$$n(\text{HCl}) \text{ in excess after NaOH addition} = 0.005 \text{ mol}$$

$$[\text{HCl}] = [\text{H}_3\text{O}^+] = 0.005 / 0.101 = 0.0495 \text{ mol L}^{-1}$$

$$\text{Final pH} = 1.31$$

- (b) Calculate the  $K_a$  of the weak acid.

**1**

$$\text{p}K_a = 4.76 \text{ (when } [\text{HA}] = [\text{A}^-], \text{ pH} = \text{p}K_a)$$

$$K_a = 10^{-4.76} = 1.7 \times 10^{-5}$$

**Question continues on next page.**

**Question 28 continued.****Marks**

When 1.00 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide solution is added to 100 mL of water, the pH increases from 7.00 to 12.00.

- (c) Explain why adding 1.00 mL of 1.00 mol L<sup>-1</sup> sodium hydroxide solution to both solution X and solution Y does not change pH of either appreciably.

**4****Solution X**

1.00 mL × 1.00 mol L<sup>-1</sup> = 1.00 mmol of NaOH is being added to both solutions.

Solution X contains 100 mL × 0.0600 mol L<sup>-1</sup> = 6.00 mmol HCl.

After reaction with the added NaOH, 5.00 mmol of HCl remains, **so it is still present in relatively large excess**, so the pH will not increase appreciably.

| Criteria                                                  | Marks |
|-----------------------------------------------------------|-------|
| Use data to show, AND state, that HCl is in large excess. | 2     |
| Use data to show, OR state, that HCl is in large excess   | 1     |

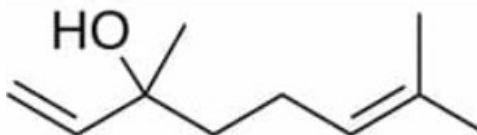
**Solution Y**

Contains 100 mL × 0.100 mol L<sup>-1</sup> = 10.0 mmol of the weak acid and 10.0 mmol of its conjugate base. Adding 1.00 mmol of NaOH will turn 1.00 mmol of the acid form into the base form, leaving 9.00 mmol of the weak acid and 11.00 mmol of the conjugate base in the resulting solution. As the ratio of conjugate base to acid is still close to 1, the pH of the solution will be close to the pK<sub>a</sub>, so the pH will not increase appreciably. Or buffer explanation.

| Criteria                                                         | Marks |
|------------------------------------------------------------------|-------|
| Use data to show, AND state, that solution is acting as a buffer | 2     |
| Use data to show, OR state, that solution is acting as a buffer  | 1     |

**Question 29** (5 marks)**Marks**

Linalool is a compound found in many flowers and spices. The structural formula is shown below.



- (a) State the names of the **two** functional groups in linalool.

**2**

1 mark: Alkene / carbon-carbon double bond

1 mark: alcohol / hydroxyl group

- (b) Linalool undergoes many chemical reactions. Complete the table by identifying the product when the following reagents are added to a linalool solution under the stated conditions.

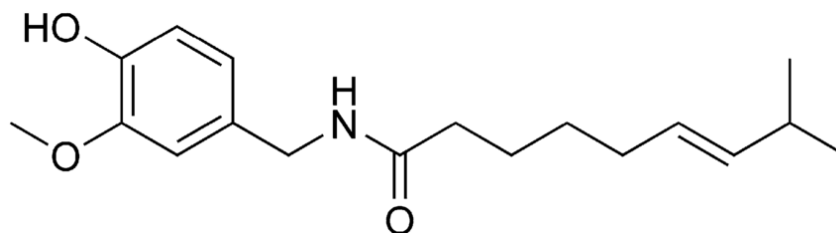
**3**

| Reagent and conditions                                                             | Structural formula of organic product |
|------------------------------------------------------------------------------------|---------------------------------------|
| $\text{H}_2(\text{g})$<br>Pd catalyst                                              | <chem>CC(C)(O)CCC</chem>              |
| $\text{Br}_2(\text{aq})$                                                           | <chem>CC(C)(Br)CCBr</chem>            |
| $\text{HCOOH}(\text{aq})$<br>$\text{H}_2\text{SO}_4$ catalyst<br>Heat under reflux | <chem>CC(=C)CC(C)(O)CC=C</chem>       |

**Question 30** (3 marks)**Marks**

The compound responsible for the hot taste of chillies, capsaicin, is shown. Drinking full cream milk, which contains fat, is recommended after eating meals containing lots of chilli as it reduces the hot taste on the tongue.

Explain why milk is better than water in removing capsaicin from the tongue.

**3**

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <i>Provides a thorough explanation in terms of intermolecular forces, including:</i> <ul style="list-style-type: none"><li>• ID capsaicin as non-polar molecule</li><li>• ID dispersion forces between milk and capsaicin</li><li>• Higher solubility in milk than water OR forms emulsion/micelles</li><li>• Must include comparison with water</li></ul> <i>A full explanation in terms of enthalpy/entropy changes was also awarded full marks</i> | 3     |
| <i>Provides a sound explanation including:</i><br><br><i>-Includes 2 points from above</i>                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <i>Provides some relevant information including:</i><br><br><i>-Forms stronger bonds with milk</i>                                                                                                                                                                                                                                                                                                                                                    | 1     |

**Question 31** (9 marks)**Marks**

Three bottles, known to contain pentan-1-ol, pentan-3-ol and 2-methylbutan-2-ol, are missing labels.

- (a) Suggest how a student could distinguish between the three alcohols using one or more chemical tests.

**3**

| Criteria                                                                                                                                                                                                                                                                                                                                                         | Marks    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <i>Give a test or two tests AND expected results for pentan-1-ol, pentan-3-ol and 2-methylbutan-2-ol</i><br><br><i>For example:</i><br><br><i>Add Lucas reagent to all three alcohols</i><br><br><i>Tertiary alcohol turns cloudy immediately</i><br><i>Secondary alcohol turns cloudy after some time</i><br><i>Primary alcohol remains colourless solution</i> | <i>3</i> |
| <i>Give test AND at least one correct result</i>                                                                                                                                                                                                                                                                                                                 | <i>2</i> |
| <i>Give test OR a correct result</i>                                                                                                                                                                                                                                                                                                                             | <i>1</i> |

Question continues on next page.

**Question 31 continued.****Marks**

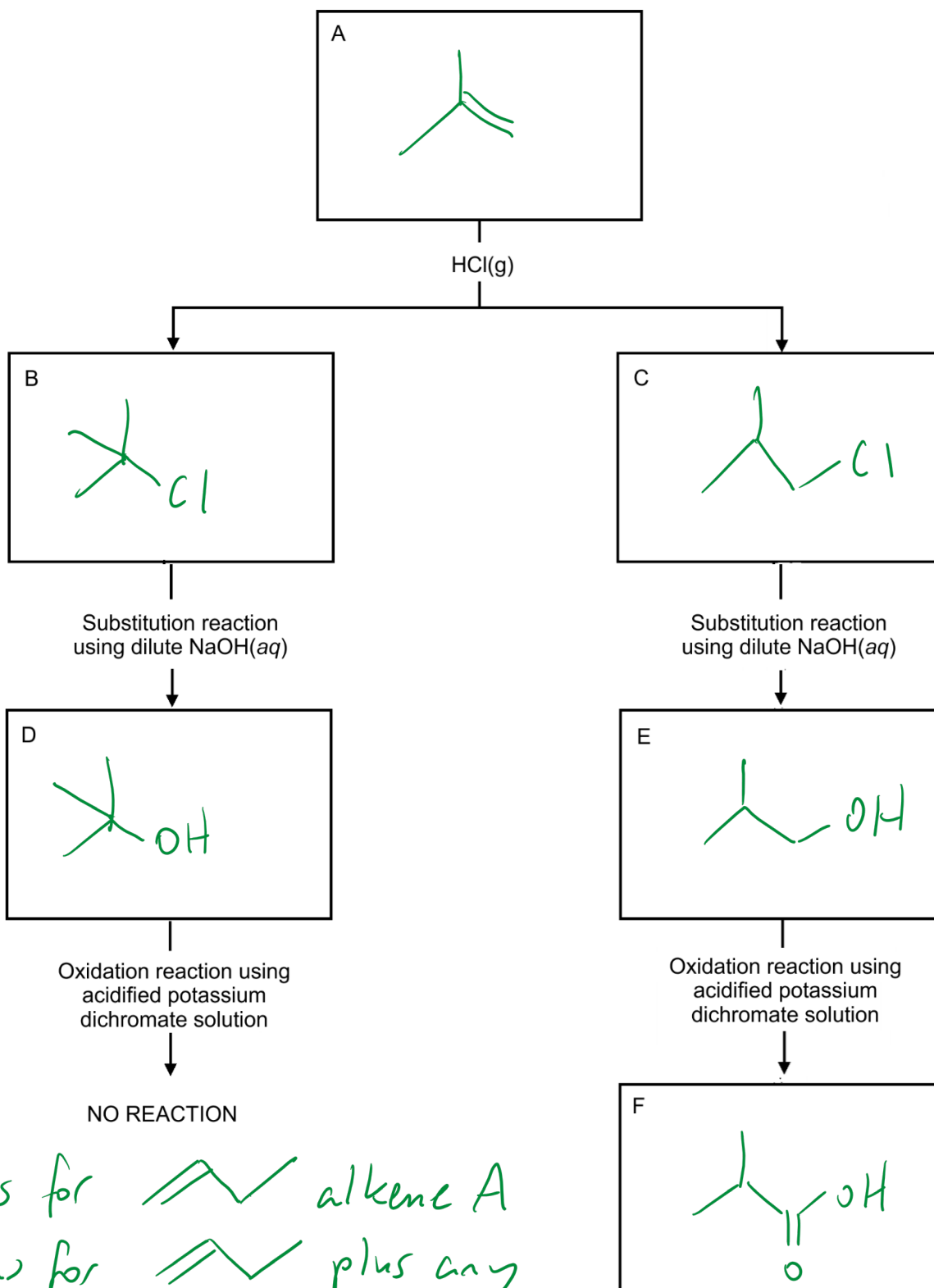
- (b) The student can use a combination of infrared, mass spectrometry and  $^{13}\text{C}$  NMR to confirm the contents of the three bottles. Evaluate the suitability of these techniques for this purpose.

**6**

| <i>Criteria</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>Marks</i> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <i>Evaluate the suitability of IR, MS and <math>^{13}\text{C}</math> NMR, by:</i><br><i>-Commenting on suitability of each method</i><br><i>-Referring to the structure/formula of the molecule</i><br><i>-Referring to the expected spectra.</i><br><br><i>IR</i> <ul style="list-style-type: none"> <li><i>Can not be used to confirm all three isomers</i></li> <li><i>Alcohols all contain O-H</i></li> <li><i>All molecules produce broad peak at 3230-3550 <math>\text{cm}^{-1}</math>.</i></li> </ul> <i>MS</i> <ul style="list-style-type: none"> <li><i>Can not be used to confirm all three isomers</i></li> <li><i>All molecules are isomers/have the same MM</i></li> <li><i>They would all produce an <math>\text{M}^+</math> peak in same position/at <math>m/z</math> 88.</i></li> </ul> <i><math>^{13}\text{C}</math> NMR</i> <ul style="list-style-type: none"> <li><i>Can confirm each individual alcohol</i></li> <li><i>Each alcohol has a different number of unique carbon environments</i></li> <li><i>Each alcohol will give a different number of peaks:</i> <ul style="list-style-type: none"> <li><i>Pentan-1-ol will give 5 peaks</i></li> <li><i>Pentan-3-ol will give 3 peaks</i></li> <li><i>2-Methylbutan-2-ol will give 4 peaks</i></li> </ul> </li> </ul> | 1) 6         |
| <i>As above, but including:</i><br><i>- 1 incomplete evaluation (missing link to structure or spectra) OR</i><br><i>- 1/2 small errors (including incorrect number of C-environments/peaks)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5            |
| <i>As above, but including:</i><br><i>- 2/3 incomplete evaluations (missing reference to spectra or structure of alcohols) OR</i><br><i>-2 full evaluations + 1 incorrect evaluation</i><br><i>-Mostly correct terminology</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4            |
| <i>As above, but with:</i><br><i>- Vague evaluations, missing links to spectra and structure of the alcohols</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3            |
| <i>Two pieces of relevant information</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2            |
| <i>One piece of relevant information</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1            |

**Question 32** (5 marks)**Marks**

Alkene A reacts with gaseous hydrogen chloride to form two isomers, B and C, with molecular formula  $C_4H_9Cl$ . A sequence of reactions for each compound is shown below. Determine the structural formulae of the compounds A to F.



3 marks for CC(C)=C alkene A

2 marks for CC(C)=C plus any error.

1 mark for any correct structure.

ACCEPT ALDEHYDE

|                         |  |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|--|
|                         |  |  |  |  |  |  |  |
| <b>CANDIDATE NUMBER</b> |  |  |  |  |  |  |  |

**Question 33** (5 marks)**Marks**

Periclase is a mineral composed of magnesium oxide ( $MM\ 40.31\ \text{g mol}^{-1}$ ). A 0.4802 g ore sample containing periclase is treated with 50.00 mL of  $0.5020\ \text{mol L}^{-1}$  hydrochloric acid. The acid reacts with the magnesium oxide, but not with the other components of the ore sample. After reaction and filtration, the solution is made up to 250.0 mL in a volumetric flask. 25.00 mL aliquots of this solution are titrated with  $0.1034\ \text{mol L}^{-1}$  sodium hydroxide (NaOH), requiring an average of 12.90 mL to reach the endpoint.

Calculate the percentage by mass of magnesium oxide in the periclase ore sample.

**5**

$$n(\text{HCl}) = 50.00\ \text{mL} \times 0.5020\ \text{mol L}^{-1} = 25.10\ \text{mmol}$$

$$n(\text{NaOH}) = 12.90\ \text{mL} \times 0.1034\ \text{mol L}^{-1} = 1.334\ \text{mmol}$$

$$n(\text{HCl in XS, in aliquot}) = 1.334\ \text{mmol}$$

$$n(\text{HCl in XS, total}) = 250.0\ \text{mL} / 25.00\ \text{mL} \times 1.334\ \text{mmol} = 13.34\ \text{mmol}$$

$$n(\text{HCl reacting with MgO}) = 25.10\ \text{mmol} - 13.34\ \text{mmol} = 11.76\ \text{mmol}$$

$$n(\text{MgO}) = 11.76\ \text{mmol} / 2 = 5.881\ \text{mmol}$$

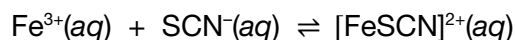
$$m(\text{MgO}) = 5.881 \times 10^{-3}\ \text{mol} \times 40.31\ \text{g mol}^{-1} = 0.2371\ \text{g}$$

$$\%(\text{MgO}) = 0.2371\ \text{g} / 0.4802\ \text{g} \times 100 = 49.37\%$$

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                    | Mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <ul style="list-style-type: none"> <li>Correctly calculates the percentage by mass of magnesium oxide in the periclase ore sample</li> </ul>                                                                                                                                                                                                                                                                | 5    |
| <ul style="list-style-type: none"> <li>Calculates the percentage by mass of magnesium oxide in the periclase ore sample, but with one error, e.g.               <ul style="list-style-type: none"> <li>Missing 1:2 mole ratio</li> <li>Missing 25 in 250 sampling factor</li> <li>Major rounding error: wrong value in 3<sup>rd</sup> s.f. (out of 4)</li> </ul> </li> </ul>                                | 4    |
| <ul style="list-style-type: none"> <li>Calculates the percentage by mass of magnesium oxide in the periclase ore sample, but with two minor errors, or one major conceptual error, e.g.               <ul style="list-style-type: none"> <li>Calculation that ignores the excess reagent / back titration concept</li> </ul> </li> </ul>                                                                    | 3    |
| <ul style="list-style-type: none"> <li>Performs two different steps of the calculation, e.g.               <ul style="list-style-type: none"> <li>Writes a balanced chemical equation</li> <li>Performs a concentration <math>\times</math> volume calculation (n.b. performing two such calculations is not sufficient for 2 marks)</li> <li>Excess/limiting reagent identification</li> </ul> </li> </ul> | 2    |
| <ul style="list-style-type: none"> <li>Provides some relevant information</li> </ul>                                                                                                                                                                                                                                                                                                                        | 1    |

**Question 34** (5 marks)**Mark**

Red iron(III) thiocyanate is produced in the following equilibrium reaction.



At equilibrium at 298 K there are 0.00540 mol of  $\text{Fe}^{3+}(\text{aq})$ , 0.00823 mol of  $\text{SCN}^{-}(\text{aq})$  and 0.0631 mol of  $[\text{FeSCN}]^{2+}(\text{aq})$  in a 100.0 mL solution.

What mass of pure sodium hydroxide should be added to increase the amount of  $\text{SCN}^{-}(\text{aq})$  to 0.00944 mol at equilibrium?

**5**

$$K_{eq} = \frac{[\text{FeSCN}^{2+}]}{[\text{Fe}^{3+}][\text{SCN}^{-}]} = \frac{\left[\frac{0.0631}{0.1}\right]}{\left[\frac{0.00540}{0.1}\right]\left[\frac{0.00823}{0.1}\right]} = 142.0$$

Let  $x$  = amount of  $\text{Fe}^{3+}$  removed by precipitation as  $\text{Fe}(\text{OH})_3(\text{s})$ .

|       | $\text{Fe}^{3+}(\text{aq}) +$                 | $\text{SCN}^{-}(\text{aq}) \rightleftharpoons$ | $\text{FeSCN}^{2+}(\text{aq})$ |
|-------|-----------------------------------------------|------------------------------------------------|--------------------------------|
| I     | 0.00540 – $x$ mol                             | 0.00823 mol                                    | 0.0631 mol                     |
| C     | + 0.00121 mol                                 | + 0.00121 mol                                  | – 0.00121 mol                  |
| E     | 0.00661 – $x$ mol                             | 0.00944 mol                                    | 0.06189 mol                    |
| Conc. | $\frac{0.00661 - x}{0.1}$ mol L <sup>-1</sup> | 0.0944 mol L <sup>-1</sup>                     | 0.6189 mol L <sup>-1</sup>     |

$$K_{eq} = \frac{[\text{FeSCN}^{2+}]}{[\text{Fe}^{3+}][\text{SCN}^{-}]} = 142.0 = \frac{[0.6189]}{[\text{Fe}^{3+}][0.0944]}$$

$$[\text{Fe}^{3+}] = 0.0462 \text{ mol L}^{-1}$$

$$\frac{0.00661 - x}{0.1} = 0.0462 \text{ mol L}^{-1}, \text{ so } x = 0.00199 \text{ mol}$$



$$n(\text{OH}^{-} \text{ added}) = 3 \times 0.00199 \text{ mol} = 0.00598 \text{ mol}$$

$$m(\text{NaOH}) = n \times \text{MM} = 0.00598 \text{ mol} \times 39.998 \text{ g mol}^{-1} = 0.239 \text{ g (to 3 s. f.)}$$

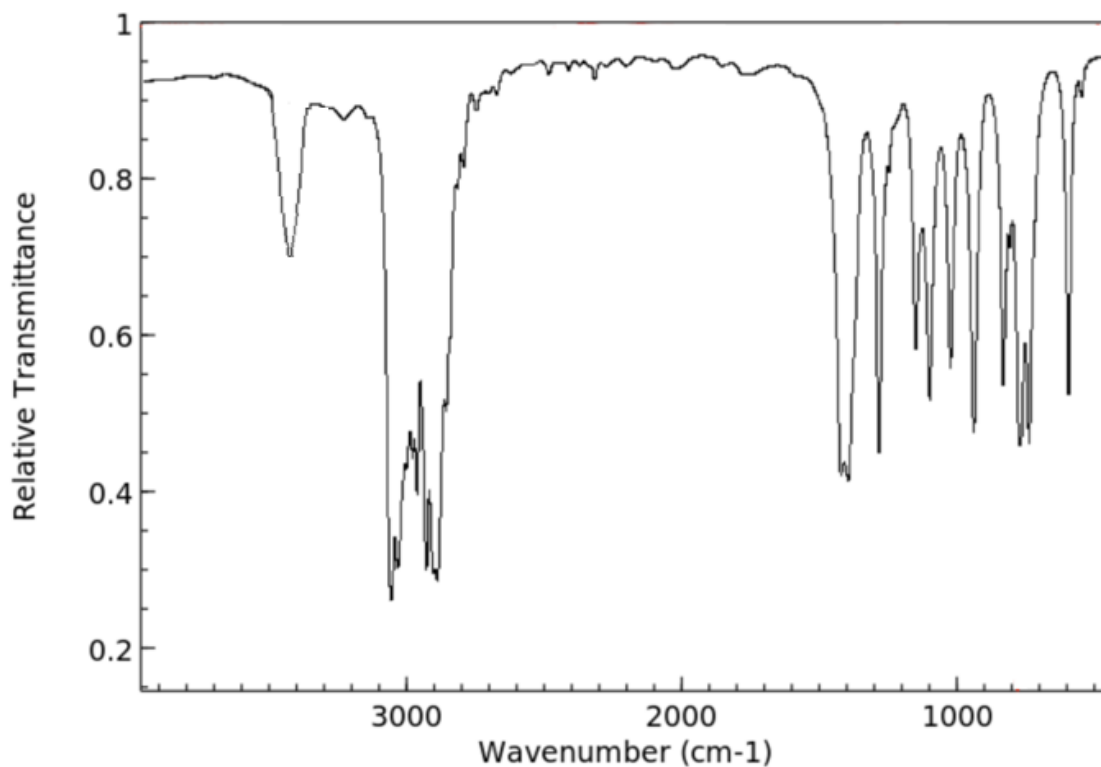
| Criteria                                                                                                                                                                                                             | Mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| • Correctly calculates the mass of sodium hydroxide added                                                                                                                                                            | 5    |
| • Performs the major steps of the calculation <ul style="list-style-type: none"> <li>◦ e.g. calculates final <math>[\text{Fe}^{3+}]</math> and/or <math>n(\text{Fe}^{3+})</math> removed</li> </ul>                  | 4    |
| • Performs some steps of the calculation <ul style="list-style-type: none"> <li>◦ e.g. calculates <math>K_{eq}</math>, <math>\Delta n(\text{SCN}^{-})</math> and substantially correct ICE box values</li> </ul>     | 3    |
| • Performs two steps of the calculation <ul style="list-style-type: none"> <li>◦ e.g. calculates <math>K_{eq}</math>, AND</li> <li>◦ <math>\Delta n(\text{SCN}^{-})</math> OR some correct ICE box values</li> </ul> | 2    |
| • Provides some relevant information                                                                                                                                                                                 | 1    |

**Question 35** (7 marks)

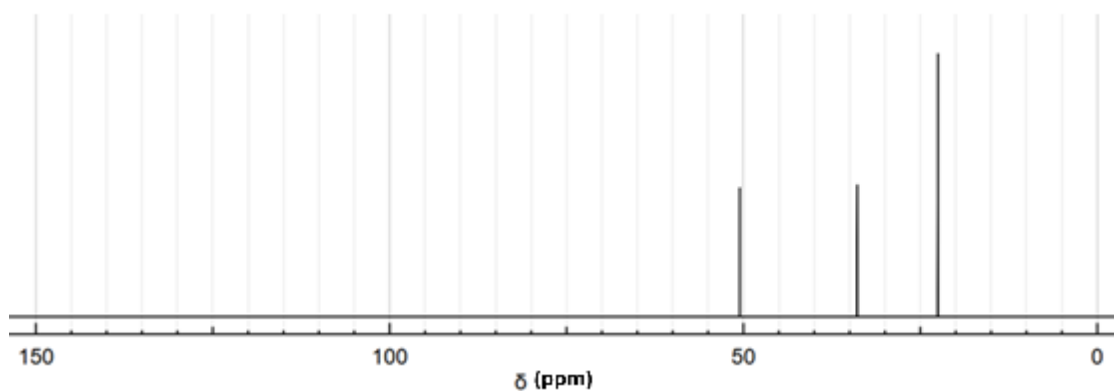
Compound Q is an organic compound which contains roughly 19% nitrogen by mass.

The infrared,  $^{13}\text{C}$  NMR and  $^1\text{H}$  NMR spectra of compound Q are shown below.

Chemical shift data for  $^1\text{H}$  NMR is provided.



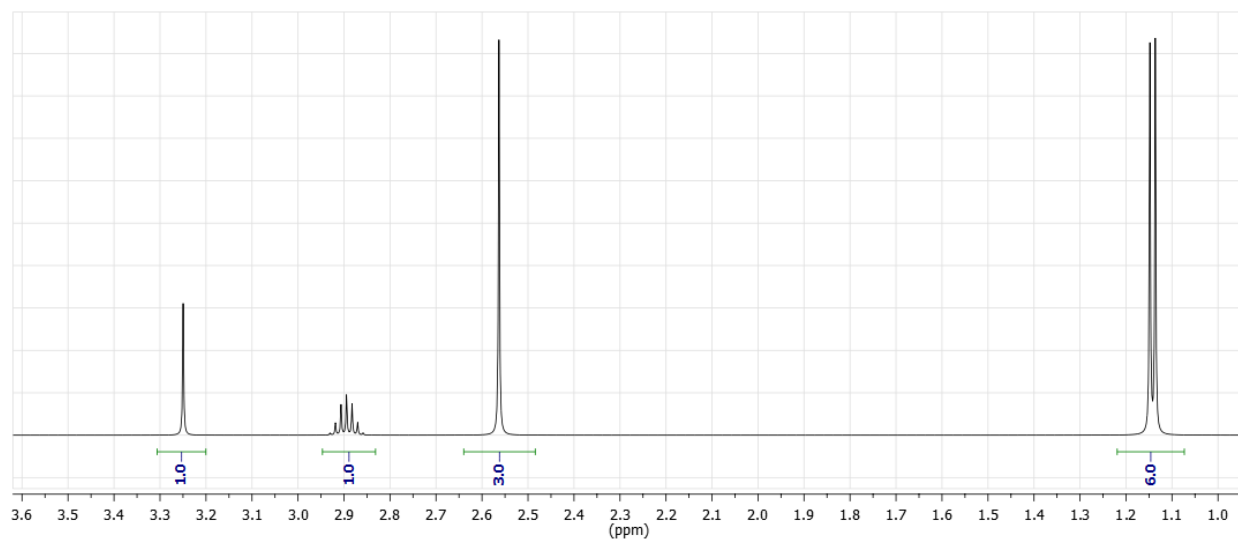
*IR spectrum of compound Q*



*$^{13}\text{C}$  NMR spectrum of compound Q*

**Question continues on next page.**

## Question 35 continued.

 $^1\text{H}$  NMR spectrum of compound Q $^1\text{H}$  NMR chemical shift data

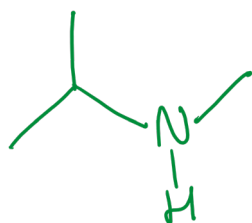
| Type of proton                                                 | $\delta/\text{ppm}$ |
|----------------------------------------------------------------|---------------------|
| $\text{Si}(\text{CH}_3)_4$ (TMS)                               | 0                   |
| $\text{R}-\text{CH}_3$                                         | 0.9–1.0             |
| $\text{R}-\text{CH}_2-\text{R}$                                | 1.2–1.5             |
| $\text{R}-\text{CHR}_2$                                        | 1.5–2.0             |
| $\text{R}-\text{C}\equiv\text{C}-\text{H}$ (alkyne)            | 2.0–3.1             |
| $-\text{CO}-\text{CH}_2-$ (aldehydes, ketones or esters)       | 2.1–2.7             |
| $\text{R}-\text{CH}_2-\text{NH}_2$ or $\text{CHR}-\text{NH}_2$ | 2.4–3.0             |
| $\text{R}-\text{CH}_2-\text{NHR}$ or $\text{CHR}-\text{NHR}$   | 2.5–3.1             |
| $-\text{CH}_2-\text{O}-$ (alcohols, ethers or esters)          | 3.3–4.8             |
| $\text{R}-\text{OH}$                                           | 1–6                 |
| $\text{R}-\text{NH}_2$                                         | 1–5                 |
| $\text{R}-\text{NHR}$                                          | 1–5                 |
| $\text{R}_2\text{C}=\text{CHR}$ (alkene)                       | 4.5–7.0             |
| $\text{R}-\text{COONH}-\text{R}$ (amide)                       | 5–9                 |
| $\text{R}-\text{CHO}$ (aldehyde)                               | 9.4–10.0            |
| $\text{R}-\text{COOH}$                                         | 9.0–13.0            |

**Question 35 continued.****Marks**

Draw and name compound Q and justify your choice using all information provided.

**7**

Compound Q

Name: **N-methylpropan-2-amine****Marking Guidelines**

| Criteria                                                                                                                                                                                                                                                                                                        | Mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <ul style="list-style-type: none"><li>• Draws and names correct structure</li><li>• Justifies structure with respect to all information provided</li></ul>                                                                                                                                                      | 7    |
| <ul style="list-style-type: none"><li>• Draws and names correct structure</li><li>• Justifies structure with respect to most information provided</li></ul> OR <ul style="list-style-type: none"><li>• Draws correct structure</li><li>• Justifies structure with respect to all information provided</li></ul> | 6    |
| <ul style="list-style-type: none"><li>• Draws structure with at least some correct parts</li><li>• Relates some information provided to structure</li></ul>                                                                                                                                                     | 4-5  |
| <ul style="list-style-type: none"><li>• Draws structure with at least some correct parts</li></ul> AND/OR <ul style="list-style-type: none"><li>• Provides some analysis of relevant information</li></ul>                                                                                                      | 2-3  |
| <ul style="list-style-type: none"><li>• Provides some relevant information</li></ul>                                                                                                                                                                                                                            | 1    |

## Sample Answer

### % composition

If 1 × N, molar mass(Q)  $\approx 100 / (19 / 14.01) \approx 74 \text{ g mol}^{-1}$

Integration of 11 × H in  $^1\text{H}$  NMR suggests 11 × H. Removing mass of  $\text{NH}_{11}$  (25) from 74 leaves 49, or approximately 4 × C.

This suggests that the formula is  $\text{C}_4\text{H}_{11}\text{N}$ , with molar mass  $73.138 \text{ g mol}^{-1}$

### IR

Peak at  $\sim 3400 \text{ cm}^{-1}$  due to N-H (possibly secondary amine)

### $^{13}\text{C}$ NMR

3 peaks (at  $\delta$  23, 35 and 50 ppm)  $\rightarrow$  some symmetry (fewer peaks than the number of carbon atoms in the molecule).

50 ppm peak definitely in C-N range (25 – 60 ppm) and 23 ppm peak must be C-C (5 – 40 ppm).

35 ppm peak is in both C-C and C-N ranges, so could be either.

### $^1\text{H}$ NMR

The 3H singlet must be a methyl group, and the chemical shift of 2.55 ppm indicates that it is attached to a nitrogen (given 2.5 – 3.1 ppm range for  $-\text{N}-\text{CH}_2-$ ):  $\text{N}-\text{CH}_3$ .

Similarly, the 1H multiplet at 2.9 ppm indicates a CH group attached to a nitrogen, with many protons on adjacent carbon atoms. The protons in the 6H doublet are split by the 1H multiplet and vice versa, which is characteristic of an isopropyl group:  $-\text{CH}(\text{CH}_3)_2$ . This gives the structure:  $\text{N}-\text{CH}(\text{CH}_3)_2$ .

The 1H singlet at 3.25 ppm must be the remaining NH proton (no splitting and its chemical shift is not distinctive (1-5 ppm, which covers all peaks present!), so not of great diagnostic use).

2019 HIGHER SCHOOL CERTIFICATE  
EXAMINATION

## Chemistry

### FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mC\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{v}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant,  $N_A$  .....  $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at  $0^\circ\text{C}$  (273.15 K) ..... 22.71 L

at  $25^\circ\text{C}$  (298.15 K) ..... 24.79 L

Gas constant .....  $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at  $25^\circ\text{C}$  (298.15 K),  $K_w$  .....  $1.0 \times 10^{-14}$

Specific heat capacity of water .....  $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

### DATA SHEET

#### Solubility constants at $25^\circ\text{C}$

| Compound             | $K_{sp}$               | Compound            | $K_{sp}$               |
|----------------------|------------------------|---------------------|------------------------|
| Barium carbonate     | $2.58 \times 10^{-9}$  | Lead(II) bromide    | $6.60 \times 10^{-6}$  |
| Barium hydroxide     | $2.55 \times 10^{-4}$  | Lead(II) chloride   | $1.70 \times 10^{-5}$  |
| Barium phosphate     | $1.3 \times 10^{-29}$  | Lead(II) iodide     | $9.8 \times 10^{-9}$   |
| Barium sulfate       | $1.08 \times 10^{-10}$ | Lead(II) carbonate  | $7.40 \times 10^{-14}$ |
| Calcium carbonate    | $3.36 \times 10^{-9}$  | Lead(II) hydroxide  | $1.43 \times 10^{-15}$ |
| Calcium hydroxide    | $5.02 \times 10^{-6}$  | Lead(II) phosphate  | $8.0 \times 10^{-43}$  |
| Calcium phosphate    | $2.07 \times 10^{-29}$ | Lead(II) sulfate    | $2.53 \times 10^{-8}$  |
| Calcium sulfate      | $4.93 \times 10^{-5}$  | Magnesium carbonate | $6.82 \times 10^{-6}$  |
| Copper(II) carbonate | $1.4 \times 10^{-10}$  | Magnesium hydroxide | $5.61 \times 10^{-12}$ |
| Copper(II) hydroxide | $2.2 \times 10^{-20}$  | Magnesium phosphate | $1.04 \times 10^{-24}$ |
| Copper(II) phosphate | $1.40 \times 10^{-37}$ | Silver bromide      | $5.35 \times 10^{-13}$ |
| Iron(II) carbonate   | $3.13 \times 10^{-11}$ | Silver chloride     | $1.77 \times 10^{-10}$ |
| Iron(II) hydroxide   | $4.87 \times 10^{-17}$ | Silver carbonate    | $8.46 \times 10^{-12}$ |
| Iron(III) hydroxide  | $2.79 \times 10^{-39}$ | Silver hydroxide    | $2.0 \times 10^{-8}$   |
| Iron(III) phosphate  | $9.91 \times 10^{-16}$ | Silver iodide       | $8.52 \times 10^{-17}$ |
|                      |                        | Silver phosphate    | $8.89 \times 10^{-17}$ |
|                      |                        | Silver sulfate      | $1.20 \times 10^{-5}$  |

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

**Infrared absorption data**

| Bond              | Wavenumber/cm <sup>-1</sup> |
|-------------------|-----------------------------|
| N—H<br>(amines)   | 3300–3500                   |
| O—H<br>(alcohols) | 3230–3550<br>(broad)        |
| C—H               | 2850–3300                   |
| O—H<br>(acids)    | 2500–3000<br>(very broad)   |
| C≡N               | 2220–2260                   |
| C=O               | 1680–1750                   |
| C=C               | 1620–1680                   |
| C—O               | 1000–1300                   |
| C—C               | 750–1100                    |

**<sup>13</sup>C NMR chemical shift data**

| Type of carbon                                                                           | δ/ppm   |
|------------------------------------------------------------------------------------------|---------|
| $\begin{array}{c}   \quad   \\ -C-C- \\   \quad   \end{array}$                           | 5–40    |
| $\begin{array}{c}   \\ R-C-Cl \text{ or } Br \\   \end{array}$                           | 10–70   |
| $\begin{array}{c}   \\ R-C-C- \\    \quad   \\ O \end{array}$                            | 20–50   |
| $\begin{array}{c}   \\ R-C-N \\   \quad \diagup \quad \diagdown \end{array}$             | 25–60   |
| $\begin{array}{c}   \\ -C-O- \\   \end{array}$ alcohols,<br>ethers or<br>esters          | 50–90   |
| $\begin{array}{c} \diagdown \quad \diagup \\ C=C \\ \diagup \quad \diagdown \end{array}$ | 90–150  |
| $R-C \equiv N$                                                                           | 110–125 |
|       | 110–160 |
| $\begin{array}{c} R-C- \\    \\ O \end{array}$ esters or<br>acids                        | 160–185 |
| $\begin{array}{c} R-C- \\    \\ O \end{array}$ aldehydes<br>or ketones                   | 190–220 |

**UV absorption***(This is not a definitive list and is approximate.)*

| Chromophore | $\lambda_{\max}$ (nm) |
|-------------|-----------------------|
| C—H         | 122                   |
| C—C         | 135                   |
| C=C         | 162                   |

| Chromophore | $\lambda_{\max}$ (nm) |
|-------------|-----------------------|
| C≡C         | 173 178<br>196 222    |
| C—Cl        | 173                   |
| C—Br        | 208                   |

**Some standard potentials**

|                                                                      |                      |                                                  |         |
|----------------------------------------------------------------------|----------------------|--------------------------------------------------|---------|
| $\text{K}^+ + \text{e}^-$                                            | $\rightleftharpoons$ | $\text{K(s)}$                                    | -2.94 V |
| $\text{Ba}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Ba(s)}$                                   | -2.91 V |
| $\text{Ca}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Ca(s)}$                                   | -2.87 V |
| $\text{Na}^+ + \text{e}^-$                                           | $\rightleftharpoons$ | $\text{Na(s)}$                                   | -2.71 V |
| $\text{Mg}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Mg(s)}$                                   | -2.36 V |
| $\text{Al}^{3+} + 3\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Al(s)}$                                   | -1.68 V |
| $\text{Mn}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Mn(s)}$                                   | -1.18 V |
| $\text{H}_2\text{O} + \text{e}^-$                                    | $\rightleftharpoons$ | $\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$  | -0.83 V |
| $\text{Zn}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Zn(s)}$                                   | -0.76 V |
| $\text{Fe}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Fe(s)}$                                   | -0.44 V |
| $\text{Ni}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Ni(s)}$                                   | -0.24 V |
| $\text{Sn}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Sn(s)}$                                   | -0.14 V |
| $\text{Pb}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Pb(s)}$                                   | -0.13 V |
| $\text{H}^+ + \text{e}^-$                                            | $\rightleftharpoons$ | $\frac{1}{2}\text{H}_2(\text{g})$                | 0.00 V  |
| $\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$                       | $\rightleftharpoons$ | $\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$   | 0.16 V  |
| $\text{Cu}^{2+} + 2\text{e}^-$                                       | $\rightleftharpoons$ | $\text{Cu(s)}$                                   | 0.34 V  |
| $\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$ | $\rightleftharpoons$ | $2\text{OH}^-$                                   | 0.40 V  |
| $\text{Cu}^+ + \text{e}^-$                                           | $\rightleftharpoons$ | $\text{Cu(s)}$                                   | 0.52 V  |
| $\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$                       | $\rightleftharpoons$ | $\text{I}^-$                                     | 0.54 V  |
| $\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$                      | $\rightleftharpoons$ | $\text{I}^-$                                     | 0.62 V  |
| $\text{Fe}^{3+} + \text{e}^-$                                        | $\rightleftharpoons$ | $\text{Fe}^{2+}$                                 | 0.77 V  |
| $\text{Ag}^+ + \text{e}^-$                                           | $\rightleftharpoons$ | $\text{Ag(s)}$                                   | 0.80 V  |
| $\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$                      | $\rightleftharpoons$ | $\text{Br}^-$                                    | 1.08 V  |
| $\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$                     | $\rightleftharpoons$ | $\text{Br}^-$                                    | 1.10 V  |
| $\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$        | $\rightleftharpoons$ | $\text{H}_2\text{O}$                             | 1.23 V  |
| $\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$                      | $\rightleftharpoons$ | $\text{Cl}^-$                                    | 1.36 V  |
| $\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$  | $\rightleftharpoons$ | $\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$ | 1.36 V  |
| $\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$                     | $\rightleftharpoons$ | $\text{Cl}^-$                                    | 1.40 V  |
| $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$                         | $\rightleftharpoons$ | $\text{Mn}^{2+} + 4\text{H}_2\text{O}$           | 1.51 V  |
| $\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$                       | $\rightleftharpoons$ | $\text{F}^-$                                     | 2.89 V  |

## PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

KEY

|                                |  |               |
|--------------------------------|--|---------------|
| Atomic Number<br>Symbol        |  | 79<br>Au      |
| Standard Atomic Weight<br>Name |  | 197.0<br>Gold |

|                               |                                |                               |                                |                              |                                 |                                |                                |                              |                                |                             |                              |                              |                                |                               |                                |                              |                              |                                |                              |                                |                            |                               |                            |
|-------------------------------|--------------------------------|-------------------------------|--------------------------------|------------------------------|---------------------------------|--------------------------------|--------------------------------|------------------------------|--------------------------------|-----------------------------|------------------------------|------------------------------|--------------------------------|-------------------------------|--------------------------------|------------------------------|------------------------------|--------------------------------|------------------------------|--------------------------------|----------------------------|-------------------------------|----------------------------|
| 1<br>H<br>1.008<br>Hydrogen   |                                |                               |                                |                              |                                 |                                |                                |                              |                                |                             |                              |                              |                                |                               |                                |                              | 2<br>He<br>4.003<br>Helium   |                                |                              |                                |                            |                               |                            |
| 3<br>Li<br>6.941<br>Lithium   | 4<br>Be<br>9.012<br>Beryllium  |                               |                                |                              |                                 |                                |                                |                              |                                |                             |                              |                              |                                |                               |                                |                              |                              | 5<br>B<br>10.81<br>Boron       | 6<br>C<br>12.01<br>Carbon    | 7<br>N<br>14.01<br>Nitrogen    | 8<br>O<br>16.00<br>Oxygen  | 9<br>F<br>19.00<br>Fluorine   | 10<br>Ne<br>20.18<br>Neon  |
| 11<br>Na<br>22.99<br>Sodium   | 12<br>Mg<br>24.31<br>Magnesium |                               |                                |                              |                                 |                                |                                |                              |                                |                             |                              |                              |                                |                               |                                |                              |                              | 13<br>Al<br>26.98<br>Aluminium | 14<br>Si<br>28.09<br>Silicon | 15<br>P<br>30.97<br>Phosphorus | 16<br>S<br>32.07<br>Sulfur | 17<br>Cl<br>35.45<br>Chlorine | 18<br>Ar<br>39.95<br>Argon |
| 19<br>K<br>39.10<br>Potassium | 20<br>Ca<br>40.08<br>Calcium   | 21<br>Sc<br>44.96<br>Scandium | 22<br>Ti<br>47.87<br>Titanium  | 23<br>V<br>50.94<br>Vanadium | 24<br>Cr<br>52.00<br>Chromium   | 25<br>Mn<br>54.94<br>Manganese | 26<br>Fe<br>55.85<br>Iron      | 27<br>Co<br>58.93<br>Cobalt  | 28<br>Ni<br>58.69<br>Nickel    | 29<br>Cu<br>63.55<br>Copper | 30<br>Zn<br>65.38<br>Zinc    | 31<br>Ga<br>69.72<br>Gallium | 32<br>Ge<br>72.64<br>Germanium | 33<br>As<br>74.92<br>Arsenic  | 34<br>Se<br>78.96<br>Selenium  | 35<br>Br<br>79.90<br>Bromine | 36<br>Kr<br>83.80<br>Krypton |                                |                              |                                |                            |                               |                            |
| 37<br>Rb<br>85.47<br>Rubidium | 38<br>Sr<br>87.61<br>Strontium | 39<br>Y<br>88.91<br>Yttrium   | 40<br>Zr<br>91.22<br>Zirconium | 41<br>Nb<br>92.91<br>Niobium | 42<br>Mo<br>95.96<br>Molybdenum | 43<br>Tc<br>Technetium         | 44<br>Ru<br>101.1<br>Ruthenium | 45<br>Rh<br>102.9<br>Rhodium | 46<br>Pd<br>106.4<br>Palladium | 47<br>Ag<br>107.9<br>Silver | 48<br>Cd<br>112.4<br>Cadmium | 49<br>In<br>114.8<br>Indium  | 50<br>Sn<br>118.7<br>Tin       | 51<br>Sb<br>121.8<br>Antimony | 52<br>Te<br>127.6<br>Tellurium | 53<br>I<br>126.9<br>Iodine   | 54<br>Xe<br>131.3<br>Xenon   |                                |                              |                                |                            |                               |                            |
| 55<br>Cs<br>132.9<br>Caesium  | 56<br>Ba<br>137.3<br>Barium    | Lanthanoids                   |                                |                              |                                 |                                |                                |                              |                                |                             |                              |                              |                                |                               |                                |                              |                              | 81<br>Tl<br>204.4<br>Thallium  | 82<br>Pb<br>207.2<br>Lead    | 83<br>Bi<br>209.0<br>Bismuth   | 84<br>Po<br>Polonium       | 85<br>At<br>Astatine          | 86<br>Rn<br>Radon          |
| 87<br>Fr<br>Francium          | 88<br>Ra<br>Radium             | Actinoids                     |                                |                              |                                 |                                |                                |                              |                                |                             |                              |                              |                                |                               |                                |                              |                              | 113<br>Nh<br>Nihonium          | 114<br>Fl<br>Flerovium       | 115<br>Mc<br>Moscovium         | 116<br>Lv<br>Livermorium   | 117<br>Ts<br>Tennessine       | 118<br>Og<br>Oganesson     |

Lanthanoids

|                                |                             |                                   |                                |                        |                               |                               |                                 |                              |                                 |                              |                             |                              |                                |                               |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|
| 57<br>La<br>138.9<br>Lanthanum | 58<br>Ce<br>140.1<br>Cerium | 59<br>Pr<br>140.9<br>Praseodymium | 60<br>Nd<br>144.2<br>Neodymium | 61<br>Pm<br>Promethium | 62<br>Sm<br>150.4<br>Samarium | 63<br>Eu<br>152.0<br>Europium | 64<br>Gd<br>157.3<br>Gadolinium | 65<br>Tb<br>158.9<br>Terbium | 66<br>Dy<br>162.5<br>Dysprosium | 67<br>Ho<br>164.9<br>Holmium | 68<br>Er<br>167.3<br>Erbium | 69<br>Tm<br>168.9<br>Thulium | 70<br>Yb<br>173.1<br>Ytterbium | 71<br>Lu<br>175.0<br>Lutetium |
|--------------------------------|-----------------------------|-----------------------------------|--------------------------------|------------------------|-------------------------------|-------------------------------|---------------------------------|------------------------------|---------------------------------|------------------------------|-----------------------------|------------------------------|--------------------------------|-------------------------------|

Actinoids

|                      |                              |                                   |                             |                       |                       |                       |                    |                       |                         |                         |                      |                          |                       |                         |
|----------------------|------------------------------|-----------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|--------------------|-----------------------|-------------------------|-------------------------|----------------------|--------------------------|-----------------------|-------------------------|
| 89<br>Ac<br>Actinium | 90<br>Th<br>232.0<br>Thorium | 91<br>Pa<br>231.0<br>Protactinium | 92<br>U<br>238.0<br>Uranium | 93<br>Np<br>Neptunium | 94<br>Pu<br>Plutonium | 95<br>Am<br>Americium | 96<br>Cm<br>Curium | 97<br>Bk<br>Berkelium | 98<br>Cf<br>Californium | 99<br>Es<br>Einsteinium | 100<br>Fm<br>Fermium | 101<br>Md<br>Mendelevium | 102<br>No<br>Nobelium | 103<br>Lr<br>Lawrencium |
|----------------------|------------------------------|-----------------------------------|-----------------------------|-----------------------|-----------------------|-----------------------|--------------------|-----------------------|-------------------------|-------------------------|----------------------|--------------------------|-----------------------|-------------------------|

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.